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Why fractions are difficult? Modeling optimal and sub-optimal integration strategies of numerators and denominators by educated adults

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ABSTRACT

Many children and educated adults experience difficulties in understanding and manipulating fractions. In this study, we argue that a major cause of this challenge is rooted in the need to integrate information from two separate informational sources (i.e., denominator and numerator) according to a normative arithmetic rule (i.e., division). We contend that in some tasks, the correct arithmetic rule is replaced by an inadequate (sub-optimal) operation (e.g., multiplication), which leads to inaccurate representation of fractions. We tested this conjecture by applying two rigorous models of information integration : (a) *functional measurement* (Experiments 1-3) and (b) *conjoint measurement* (Experiment 4-5) to data from number-to-line and comparative judgment tasks. These allowed us to compare participants' integration strategies with that of an ideal-observer model. Functional measurement analyses on data from the number-to-line task, revealed that participants could represent the global magnitude of proper and improper fractions quite accurately and combine the fractions' components according to an ideal-observer model. However, conjoint measurement analyses on data from the comparative judgment task, showed that most participants combined these fractions' components according to a sub-optimal (saturated) observer model, that is inconsistent with an ideal-observer (additive) model. These results support the view that educated adults are capable of extracting multiple types of representations of fractions depending on the task at-hand. These representations can be either accurate and conform with normative arithmetic or approximated and inconsistent with normative arithmetic. The latter may lead to the observed difficulties people experience with fractions.

1. Introduction

Fractions are ubiquitous in our daily life. They serve as an important means for conveying quantitative aspects of reality such as: ratios, percentages, proportions, part-whole relations etc. Knowledge of fractions is related to the acquisition of more sophisticated mathematical skills (Bailey, Hoard, Nugent, & Geary, 2012; Siegler et al., 2012), and is therefore crucial for occupational and economic success (Davidson, 2012; McCloskey, 2007). People who mastered fractions at an early stage of life also show better achievement in mathematics at later stages (Fazio, DeWolf, & Siegler, 2016). However, many children (Barbieri, Rodrigues, Dyson, & Jordan, 2020; Braithwaite & Siegler, 2021; Bright, Behr, Post, & Wachsmuth, 1988; Hartnett & Gelman, 1998; Lee & Lee, 2023; Siegler, Thompson, & Schneider, 2011) and adults (McCloskey, 2007) are challenged by fractions. A plausible reason is that fractions require a developmental trajectory from experience with whole numbers to deeper understanding of other types of numbers (Siegler, Fazio, Bailey, & Zhou, 2013; Vamvakoussi & Vosniadou, 2004). This entails a shift from dealing with uni-dimensional integers to two-dimensional objects composed of a numerator and a denominator (DeWolf, Bassok, & Holyoak, 2015). To perceive the global magnitude of a fraction, one must combine the information from the two dimensions, or alternatively evoke (if exists) a corresponding holistic representation, which also must reflect the correct relations between denominator and numerator.

An important question that attracted considerable attention from researchers concerned whether symbolic fractions are represented holistically as integrated wholes along a mental number line (Schneider & Siegler, 2010), or, alternatively, relay on analytic processing of their components (i.e., denominator and numerator) (Bonato, Fabbri, Umilta, & Zorzi, 2007). We argue that this question cannot fully explain why people are challenged by fractions, and therefore should be augmented or replaced by another question concerning the precision of the internal representation, and the psychological composition rules (e.g., division) by which numerators and denominators are combined to produce global magnitude. Note that imprecise or inaccurate representations can exist with either holistic or analytic representations. The current

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study examines the precision of the internal representations themselves as a function of the accuracy of the integration process that leads to those representations. In particular, we uncover the underlying psychological *composition rules* (Luce & Tukey, 1964) by which denominators and numerators are integrated. How do educated adults combine the components of a symbolic fraction to arrive at its global magnitude? Can people apply optimal integration rules? When do they fail to do so? Are 1/2 and 2/4 represented the same way? To-date, these questions have been rarely examined; maybe because most current theories in numerical cognition are theories of whole number representation (but see, Siegler et al., 2011). Take for example the set of stimuli that is often recruited by the great majority of studies; these use fractions of different magnitudes (e.g., 1/2, 1/3), but rarely do they present fractions of the same magnitude that are composed of different components (e.g., 1/2 and 2/4). We contend that investigation of fractions requires more sophisticated methodologies than those applied with integers.

The theoretical caveats presented here are crucial for understanding the difficulties people experience when dealing with fractions. In this study we test competing models of integration and compare them with those of an *ideal observer* – an observer whose performance reflects the correct solution – in order to estimate the efficiency by which such computational processes are performed. En-route to this goal, we address related issues, such as the underlying psychological scales of fractions (i.e., linear vs. logarithmic), and the precision of estimation and representation. Our models build on classic psychophysical measurement ideas such as *conjoint measurement* (Krantz, Luce, Suppes, & Tversky, 1971; Luce & Tukey, 1964) and *functional measurement* (Anderson, 1981), which are applied here to data from traditional non-speeded comparative judgments (Fitousi, 2010; Moyer & Landauer, 1967) and number-to-line (Rips, 2013) tasks performed by educated adults. The current work sheds new light on the mental alchemy by which transformation takes place from whole numbers magnitudes to fraction magnitudes.

2. Why fractions are difficult?

Learners often find fractions to be a difficult topic. In their "integrated theory of numerical development", Siegler et al. (2011) ascribed this difficulty to conceptual discontinuities between whole numbers and fractions. Whole numbers are: (a) written as a single symbol, (b) have unique successors, (c) are countable, and (d) never increase by division. Fractions, in contrast, have none of these properties. However, children and adults who have not learned fractions to a sufficient degree of mastering, assume that properties of whole numbers are properties of all other types of numbers. This leads to such errors as assuming that 1/4 is larger than 1/3. Thus, successful learning of fractions requires a complete reorganization of numerical knowledge. It also requires the understanding that fractions, like whole numbers, represent magnitudes that can be located on a number line (Siegler et al., 2013). Another source of difficulty is that due to education, children and adults often view fractions in terms of part/whole relations which leads to misunderstanding of improper fractions (e.g., 5/4). This robust tendency to use single-unit counting to interpret fractions and the difficulties that arise due to this tendency has been dubbed the "whole number bias" (Ni & Zhou, 2005). Early preschool and elementary education is dominated by relations and operations of integers, and thus mastering of fraction requires a conceptual change (Hartnett & Gelman, 1998).

Knowledge of symbolic fractions develops later than that of non-symbolic knowledge (Goswami, 1989). Kindergarten children exhibit some understanding of the fraction 1/2, but only at the age of seven do they understand the inverse relations between the value of objects received (numerator) and sharers (denominator) (Sophian, Garyantes, & Chang, 1997). The written notation of fractions is introduced at 4th grade and pose real challenges to learners. Improvement is observed, but it is slow (Siegler & Pyke, 2013). The difficulties experienced by children in acquisition of rational numbers may continue to affect their performance in their adulthood (Bailey et al., 2012). Fractions and proportions are difficult even for educated adults (Gigerenzer & Hoffrage, 1995). Several studies have shown that 8th graders and college students could not achieve accuracy levels higher than 70% correct in comparison problems (Schneider & Siegler, 2010; Siegler et al., 2011). The difficulty may stem from people's natural tendency to handle natural frequencies rather than proportions (Matthews, Lewis, & Hubbard, 2016). With symbolic fractions the difficulties may also result from the notation by which fractions are often written. Fractions are multidimensional objects that are composed of two integers often written one over the other with a line separating them (DeWolf et al., 2015). Unlike whole numbers, the magnitude of the number is not immediately discernible. The global magnitude should be retained from the constituting integer components (numerator and denominator). In this sense, fractions are similar to two-digit numbers (Dehaene, Dupoux, & Mehler, 1990; Fitousi & Algom, 2020). In both cases, the global meaning of the number is discerned according to mathematical laws from individual elements. Treating each component separately may hamper performance, as exemplified by the whole integer bias (Bonato et al., 2007; DeWolf et al., 2015).

3. From integers to fractions

Numerical cognition researchers assume that symbolic and non-symbolic natural numbers are represented as analog quantities a long a mental number line (Ansari, 2008; Dehaene, 2011). According to this idea numerals, whenever presented for any purpose, are transformed automatically onto a magnitude representation along an analog number line (Dehaene, Dehaene-Lambertz, & Cohen, 1998; Gallistel & Gelman, 2000). The number-line metaphor has been supported by a great deal of behavioral, neuroimaging and modeling research showing that both symbolic and non-symbolic quantities are subjected to similar behavioral and neuronal phenomena (Halberda & Feigenson, 2008). For example, in the well-known comparative judgment task, decision time and error rates decrease as a function of numerical distance (Moyer & Landauer, 1967). This *distance effect* has been observed with two-digit numbers (Dehaene et al., 1990; Fitousi & Algom, 2020) and symbolic fractions (Schneider & Siegler, 2010). The finding of a distance effect has been taken to support the notion of a holistic integrated representation (but see, Nuerk, Weger, & Willmes, 2001). Another feature of the number-line is that it is spatially oriented. Evidence from the SNARC (spatial-numerical association of response codes) effect suggests that large numbers are responded faster with a right-hand than a left-hand key, while the reverse holds true for small numbers (Dehaene, Bossini, & Giraux, 1993; Fitousi, Shaki, & Algom, 2009).

The psychophysical function that relates actual (objective) numerical magnitude to psychological (internal) magnitude has also been the subject of extensive investigations (Fitousi, 2022; Siegler & Opfer, 2003). Research with integers yielded two alternative functions. According to the Fechnerian approach, numbers are represented logarithmically such that "equal space is allocated to the interval between 1 and 2, between 2 and 4, and between 4 and 8" (Dehaene, 2011, p.73). This approach has been supported by the presence of distance and size effects in comparative judgment tasks with symbolic and non-symbolic numbers. In contrast, according to the accumulation model (Gibbon & Church, 1981) quantities, including numbers, are represented linearly with increasing scalar variability (Gallistel & Gelman, 2000). This approach can explain the distance and size effect. However, using number-to-line tasks, Siegler and Opfer (2003) have shown that children start with a logarithmic representation, but as they grow up they also acquire a scalar representation. Thus, both types of representations exist in educated adults.

Research in the domains of fractions has been guided by the same behavioral (Bonato et al., 2007), single cell recording (Nieder, Freedman, & Miller, 2002), and neuroimaging (Ischebeck, Schocke, & Delazer, 2009) methods which were deployed to study integers. A central question in all of these studies concerned the nature of internal

representation that guide the understanding and manipulation of fractions. In particular, researchers asked whether fractions are mapped onto the same number-line as integers. Two diametrically opposed theoretical alternatives have been offered. According to the *holistic* approach (Schneider & Siegler, 2010), fractions are transformed into an analog magnitude representation along a mental number line just like integers. Thus, this representation can be readily accessed, without the need to retrieve the constituent components (i.e., denominator and numerator). Support in this conclusion has been given primarily by the findings of a logarithmic distance effect in comparative judgments (DeWolf, Grounds, Bassok, & Holyoak, 2014). In contrast, the *analytic* approach (Bonato et al., 2007; Kallai & Tzelgov, 2009) postulates that fractions have no unique analog representation, but are accessed via their components (numerator and denominator). The evidence for this conclusion came from the absence of *size congruity effects* in fraction–fraction pairs (Kallai & Tzelgov, 2009), and the absence of a distance effect (Bonato et al., 2007).

4. Fractions: Pinpointing the challenge

It has been often argued that people's difficulties with fractions stems from the lack of holistic representation (Bonato et al., 2007; Kallai & Tzelgov, 2009). We contend that in order to understand why people are challenged by fractions, one should augment or even replace the holistic/analytic question with another question regarding the precision of representation and the composition rule by which numerators and denominators are combined. People can err whether they represent fractions holistically or analytically, thus the question of holistic versus analytic processing is secondary to the question of precision and integration. For example, in this study we show formally that some models of holistic and analytic processing of fractions are indistinguishable. Moreover, in comparative judgments as well as in other tasks (e.g., number-to-line), people may use rules of thumbs and fast and frugal strategies (Gigerenzer & Todd, 1999) that do not necessarily reflect an accurate and precise representation. For example, people can approximate the magnitude of large numbers without representing their exact magnitudes. These strategies can still lead to "correct" responses. Generally, fast and frugal strategies are known to produce successful judgments in some tasks but not in others (Gigerenzer, Czerlinski, & Martignon, 1999). For example, Kallai and Tzelgov (2009) have concluded that fractions are represented in long-term memory only vaguely as "entities smaller than 1".

We therefore contend that the question of holistic/analytic representation is secondary to the question of how the components are represented and integrated. For example, none of the reviewed studies tested the question of whether 1/2, 2/4 and 3/6 are processed in the same fashion (but see, Braithwaite & Siegler, 2018). The great majority of studies recruited fractions based on magnitude value but not based on their componential structure. Such set of stimuli cannot inform us on the way the information from the denominator and numerator has been integrated. To answer this question, one has to employ factorial designs of stimuli, and apply well-suited techniques. The upshot is that accuracy may or may not necessarily reflect accurate representation. What is more important to know is how precise the representations are. Unlike previous studies, we highlight the fact that people's response to fractions reflect an integration processes of denominators and numerators. It is the result of this combination process that determines whether responses are correct or incorrect, irrespective of whether the representation is holistic or not. To-date, this process of integration has not been studied in detail. It is the goal of this study to uncover it.

The critical question in this study is whether fractions' components are integrated according to the correct mathematical rule or not. And if the latter, what is the actual operative rule. Ideally, acquisition of an accurate "fraction sense" should result in optimal performance, not only in comparative judgments, but also in other tasks as well (e.g., number-to-line). However, we know from research on integers that humans may acquire, not one, but various types of representations for the same mathematical object and apply a different representation on each occasion depending on the task (Siegler & Opfer, 2003; Siegler, Thompson, & Opfer, 2009). Thus, it is important to assess the integration process and the attendant combination rule under various tasks and contexts, as we do here. One notable context is the format by which numbers are represented (Arabic digits, number words). Research on single digit numbers has revealed significant differences in the processing of number-words and Arabic digits (Damian, 2004; Sullivan & Barner, 2013). In addition, several studies have demonstrated the effect of language on the way fractions are understood (Miura, Okamoto, Vlahovic-Stetic, Kim, & Han, 1999; Paik & Mix, 2003). We address this issue in Experiment 2.

5. Symbolic fractions and composition rules

We are interested in how the internal representation of the global magnitude of a fraction changes as a function of the internal representations of denominator and numerator values. Many scientific projects can be viewed as an attempt to explain changes in some dependent variable as a function of variations in one or more independent variables. Take for example Newton's second law that $F = m \times a$. The force (F) that is acting on an object is equal to the product of its mass (m) and its acceleration (a). To verify the law, one has to measure F, m, and a independently, and then show by calculation, that the law holds. The situation in Psychology is complicated, because we do not have direct access to the internal representations of the pertinent psychological variables. However, psychologists have formulated ingenious techniques based on composition rules to describe functional relations among various variables (Anderson, 1981; Luce & Tukey, 1964). For example Anderson (1971b) developed a *functional measurement* technique which is an algebraic judgment model that describes the psychological combination rules by which two sources of information (variables) are combined into a single uni-variate judgment. The model specifies both the scale and weight values of each piece of information, and the mathematical operation by which they are composed. Functional measurement postulates the existence of three universal laws of integration (addition, multiplication and averaging). It is important to note that what is integrated in functional measurement is the psychological valuation of the components, and not the physical or actual values. Stimuli that do not undergo a valuation process of the constituent dimensions lie outside the domain of functional measurement. Integration does not exist with these dimensions because there are no psychological values to integrate.

Functional measurement has been applied to a large scope of psychological judgments (Anderson, 1981). A study by Ben-Nathan and Algom (2007) used functional measurement to understand whether the components of two-digit numbers undergo a valuation and integration processes, or alternatively, perceived holistically as composed objects from the start. Participants placed two-digit numbers on a 0–100 line. The authors displayed the responses (length in cm from the left-end of the line) in a factorial plot. The units were placed on the abscissa, with a separate curve for each of the decades. The valuation–integration hypothesis predicted that the data should form a linear fan, which reflects an underlying adding rule of integration that is operating on changing scale values. The interaction is due to the operation of Weber's law. The alternative hypothesis predicted a set of parallel curves, reflecting the normative rule by which a long-term representation has been acquired. The authors have found parallel lines, which supported the second hypothesis, namely that two-digit numbers are not subjected to a valuation and integration processes, but are represented holistically with subjective magnitude being governed by a strictly additive composition rule, that is consistent with the normative mathematical model. An absence of interaction provided strong statistical evidence

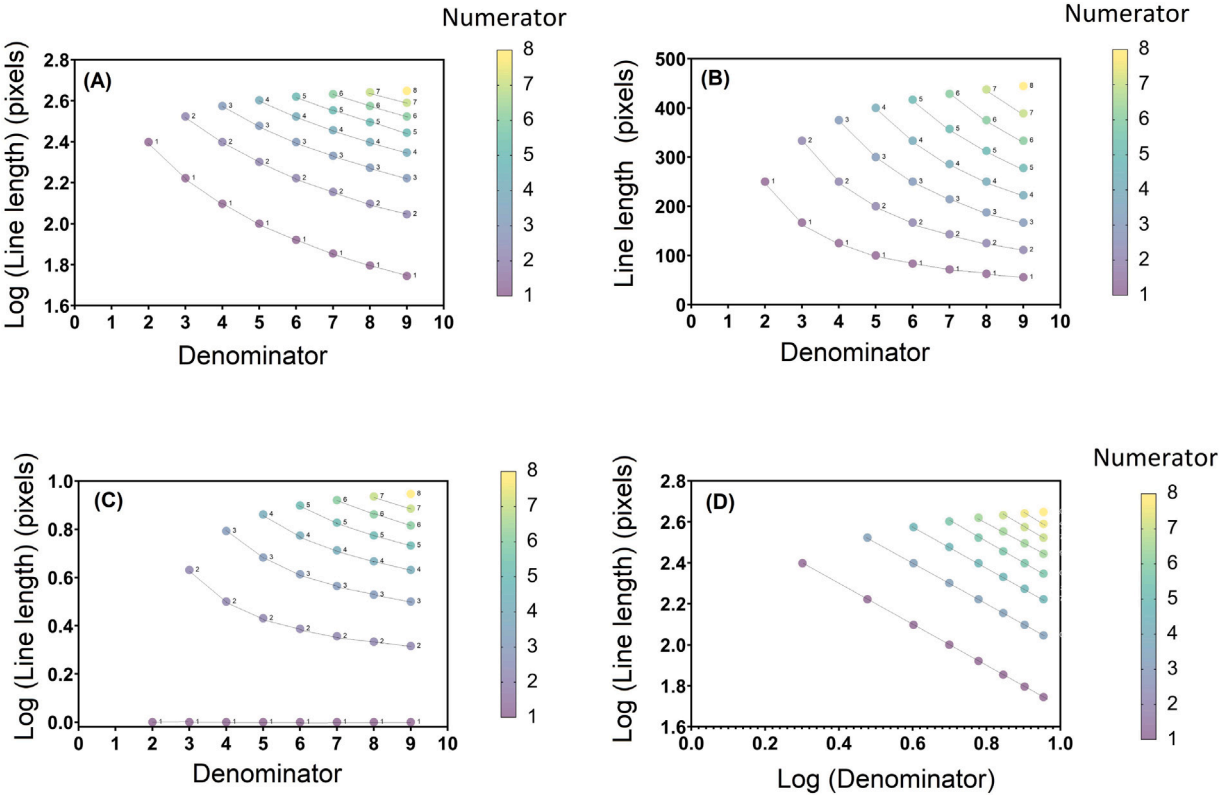


Fig. 1. Four hypothetical functional measurement models for the perception of proper fractions: (a) holistic-logarithmic, (b) holistic-linear or analytic-linear, (c) analytic-logarithmic, and (d) log–log representation of (b). See Text for details.

for a linear representation of single- and two-digit numbers along a mental number line.[1]

A related technique dubbed *conjoint measurement*[2] was developed by Luce and Tukey (1964). The approach and its subsequent developments – *Maximum Likelihood Conjoint Measurement* (MLCM Ho, Landy, & Maloney, 2008; Knoblauch & Maloney, 2012), allows researchers to simultaneously derive the psychological scales for each dimension (in ratio scale) along with the composition rule, based on data from comparative judgments (in ordinal scale). Three major models of integration can be tested in MLCM: (a) an independent observer model, (b) an additive observer model, and (c) a saturated observer model. The MLCM has been applied to a gamut of physical dimensions from: hue, chroma and lightness (Gerardin, Dojat, Knoblauch, & Devinck, 2018; Rogers, Knoblauch, & Franklin, 2016), to surface gloss and bumpiness (Ho et al., 2008; Qi, Chantler, Siebert, & Dong, 2015), to time (Lisi & Gorea, 2016), facial dimensions (Abbatecola, Gerardin, Beneyton, Kennedy, & Knoblauch, 2021; Nichiporuk, Knoblauch, Abbatecola, & Shevell, 2018) and numerical dimensions (Fitousi, 2022).

[1] Please note that the "mental number line" is a theoretical concept that relates to the idea that numbers are represented spatially along an internal left-to-right line, such that small numbers appear on the left-hand side and large numbers appear on the right-hand side. However, the term "number-to-line" refers to a task in which participants are asked to draw a line that corresponds to the perceived magnitude of a number.

[2] Functional measurement and conjoint measurement can be considered as complimentary approaches. However, historically, they were developed separately, and reflected very different (even rival) theoretical view points, as can be noted in the exchange between the authors of these two approaches (see, Anderson, 1971a; Krantz & Tversky, 1971).

6. Research strategy and predictions

The current study applies information integration models (Anderson, 1981; Luce & Tukey, 1964) to symbolic fractions. Our goal is to understand how numerators and denominators are represented, and how they are combined into global magnitudes. Here, the perception of a fraction is viewed as a single judgment that is affected by variations in the values of denominator and numerator. Do people integrate the components in an optimal (according to the normative mathematical operation) or in a sub-optimal way (e.g., deviate from the normative mathematical operation)? Do the components affect the manner by which the whole number is represented. Is 3/6 represented the same way 4/8? What are the psychological rules (e.g., division) that guide people judgments of fractions? It is important to note that according to information integration models, the integration process is not held with the actual physical values, but rather with the psychological representations of these values. Thus, the goal is not only to model the integration process, but also to uncover the internal representations on which these processes are held.

In Experiments 1, 2 and 3, we have deployed a *number-to-line task*, in which participants were presented with a single symbolic fraction, written either in Arabic notation, or as a Hebrew word/s. Participants were asked to draw a corresponding line on a 0–1 or 0–10 scale. We applied functional measurement (Anderson, 1981) to these data. In Experiment 4 and Experiment 5, we deployed a non-speeded *comparative judgment* task (Rinne, Ye, & Jordan, 2017). Participants were presented with a pair of symbolic fractions written in Arabic notation. These appeared either in a sequential or parallel mode of presentation. Data from those experiments were analyzed with the MLCM technique (Ho et al., 2008; Knoblauch & Maloney, 2012).

7. Experiments 1–3: Functional measurement

We adopted the logic of functional measurement (Anderson, 1981) to derive various observer models of symbolic fractions. This was

accomplished by displaying hypothetical responses (length in pixels from the left-end of the line) in a factorial plot. Experiments 1–2 are semi-factorial designs because they consists of the values 2–9 for denominator and 1–8 for numerator. Combining all of these values resulted in all possible proper fractions (see Fig. 2). We projected the expected line length for each proper fraction on the y axis, while the denominator values were placed on the abscissa, with a separate curve for each of the numerator values. The first model is dubbed the *holistic-logarithmic* observer. According to this model, the representation is holistic and the global fraction magnitude is converted onto a logarithmic scale. Thus, subjective magnitude should be proportional to the log-transform of the objective magnitude:

$$Mag \propto \log(\frac{X}{Y}) \tag{1}$$

where x and y are numerator and denominator, respectively. The resulting factorial plot is a family of decreasing functions (see Fig. 1a). The fan pattern is consistent with the idea that the overall fraction's magnitude decreases as the denominator increases, and the rate of decrease is different for each level of numerator. A second model is called the *holistic-linear* observer, and it is based on the assumptions that fractions are represented as wholes, with the global magnitude represented in a linear scale. In this case the following should hold:

$$Mag \propto (\frac{X}{Y}). \tag{2}$$

Fig. 1b presents the expected functional measurement model, which consists of a family of decreasing functions. As can be noted, the model yields quite similar functions, though not identical, to those produced by the holistic-linear. One notable difference that can assist us in dissociating the models when tested against empirical data is the behavior of the curves for the fractions 1/9 and 2/9. The distance between the curves for numerators 1 and 2 is constant in the holistic-logarithmic model, but decreasing in the holistic-linear model.

The third model is called *analytic-logarithmic* observer. It is based on the premises that the representation of fractions is decomposed, and that components are subjected to logarithmic transformation. In that case, the construction is accomplished by dividing the internal transformed values. The resulting subjective magnitude amounts to:

$$Mag \propto \frac{\log X}{\log Y}. \tag{3}$$

Fig. 1c presents the functional measurement realization of this model. As can be noted, the model produces a family of decreasing functions, except for cases in which the numerator is 1. This occurs because $log(1) = 0$. In psychological reality this might be solved by adding a small constant.

Finally, a fourth *analytic-linear* observer model entails a representation that is decomposed, but with the components being converted according to a linear scale. The functional measurement realization of this model is depicted by the same Eq. (1) as the *holistic-linear* model. Thus, the two models are mathematically indistinguishable (see Fig. 1b). This strengthen our claims that the question of holistic/analytic representation cannot fully explain why fractions are difficult, because both can yield comparable behavioral results, that are mathematically indistinguishable. In Experiment 3 we used a full-factorial design, that resulted in both proper and improper fractions. Similar models were derived for these set of fractions. A comparable analysis is performed with respect to non-speeded comparative judgments of fractions, using the MLCM approach (see Section: Experiment 4–5: Maximum Likelihood Conjoint Measurement).

It should be noted that each of the four possible models also determines the shape of the psychophysical function that relates the fraction's global magnitude to its estimated line-length. The holistic-logarithmic and analytic-logarithmic predict a non-linear function, whereas the holistic-linear and analytic-linear models predict a linear function.

Another point that deserves a comment at this stage concerns the nature of combined representations and their relations to mathematical operations. A fundamental axiom in *foundation of measurement* (Krantz et al., 1971; Luce & Tukey, 1964) states that conjoint psychological representations are consistent with any additive operation (e.g., addition, subtraction). This can be readily demonstrated in the case of fractions. Although the normative composition rule is division, we can show that by taking the logarithm of a fraction,[3] its global magnitude can be expressed in terms of an addition/subtraction operation:

$$\log(\frac{x}{y}) = \log x - \log y. \tag{4}$$

The resulting functional measurement model is a family of linear decreasing functions (seen Fig. 1d), which clearly entail that the components are integrated according to an additive rule. The tilted parallel lines entail separate influences of denominator and numerator values that increase/decrease additively, with no interaction between the components. This insight also provides important constraints on the composition rules expected if integration is performed according to the normative mathematical operation. In the case of fractions, any deviation from additivity in a log–log scale implies sub-optimal processing. Such deviations can occur when the numerator and denominator are integrated independently of each other, or alternatively (and more likely) when they are combined interactively. In both cases, the integration is not consistent with the normative mathematical operation (Krantz et al., 1971; Luce & Tukey, 1964).

At this point, it is important to explicate our terminology with respect to what are *normative* and *optimal* strategies. We use the two interchangeably. By normative/optimal strategy we refer to the correct mathematical rule by which the components are integrated. For example, with two-digit numbers the normative rule is additive. With fractions the normative rule is either division or, addition (in log terms). Sometimes an incorrect mathematical rule can still lead to optimal performance in a given task, but we contend that an optimal strategy is one that allows learners to excel under all tasks. Thus, we assume that the *holistic-linear* and the *analytic-linear* (which were shown to be indistinguishable) are the normative-optimal models of integration. Any combination rule that deviates from these models is sub-optimal, because it does not reflect global magnitude accurately.

8. Experiment 1

In this experiment participants were presented with proper fractions written in Arabic notation, and were asked to draw a corresponding line on a 0–1 scale. The structure of the stimuli set consisted of all possible combinations of denominator and nominator values (1–9) that create a proper fraction with a global magnitude that is smaller than 1. The denominator values spanned between 2 and 9, whereas the numerator values ranged between 1 and 8. Studies with symbolic integers have shown that there is "log-to-linear" shift in estimation (Siegler et al., 2009), according to which young children or uneducated adults produce logarithmic estimates, whereas educated adults produce linear estimates (Opfer & Siegler, 2007; Siegler & Opfer, 2003). It is still not clear whether educated adults exhibit such a shift in the domain of fractions.

9. Method

9.1. Observers

A total of 30 observers participated in this study. They were recruited from Ariel University pool of participants (Females = 24, Males

[3] The logarithmic transformation here has nothing to do with Weber's law in numerical cognition.

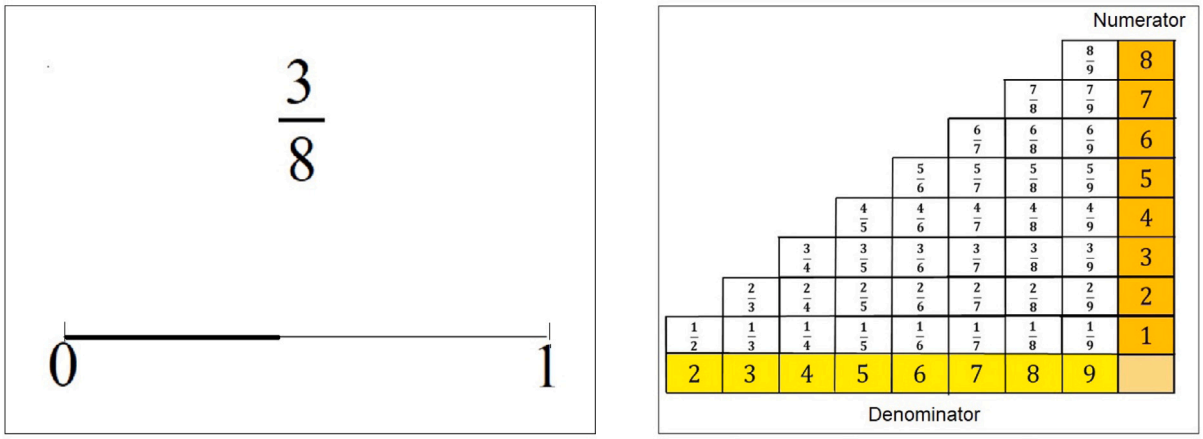


Fig. 2. Experiment 1: Left Panel: An example for a number-to-line trial. Right-Panel: The proper fractions in the experiment were constructed by combining each possible numerator values (1–8) with each possible denominator values (2–9).

= 6, mean age = 23.5, *sd* = 2.98). These participants were Psychology students of medium-to-high socioeconomic status. They were compensated with a course credit as part of an introductory course. All observers had normal or corrected-to-normal vision. All observers were naive to the goals of the study. The samples in Experiment 1 - 3 included each 30 participants based on previous studies (Sullivan, Juhasz, Slattery, & Barth, 2011).

9.2. Stimuli and apparatus

The stimuli were fractions presented as black characters over a white background. Each fraction subtended 31.7 mm in height and 12.7 mm in width. The denominator values included the numbers 2–9, while the numerator values included the numbers 1–8. Combining all possible denominator and numerator values resulted in 36 proper fractions (i.e., magnitude <1). The fractions were separated by a horizontal bar. Fig. 2 presents the stimuli set on a Cartesian grid with the denominator and numerator serving as axes. The number line was presented below the fraction with the numbers 0 and 1 posited at the left- and right- hand sides of it, respectively (see Fig. 2). The line subtended 500 pixels. All numerical characters were presented in font type 'New Times Roman' and in size 48. Viewing distance was approximately 70 cm.

9.3. Procedure and design

The experiment was programmed in Macromedia Authorware 7.0 software, and was run on an ASUS N56VM-S4041V laptop computer with a 15.6 inch screen and 1366 × 768 resolution. Each block in the Experiment consisted of 36 trials. There were eleven blocks in the Experiment, so the total number of trials amounted to 396. On each trial one of the fractions was randomly chosen by the computer (without replacement) and presented until response. A symbolic number-line was placed below the fraction. It was marked by a horizontal thin line with the digits '0' and '1' placed on its left-hand and right-hand sides, respectively. Participants were asked to estimate the overall magnitude of the fraction and produce a line that expressed their subjective estimation, starting from the left-hand side of the number-line and ending somewhere on the right-hand side of the line, according to their subjective estimation. The generated line was a black thick line which covered the estimated proportion of the thin number line. It was produced by the participants by clicking the mouse and moving it along the so called number line. Once the participant was satisfied with their estimated line, they could remove their finger from the mouse, and the line length was placed. At this stage its length in pixels was recorded by the computer. To move on to the next trial participants pressed the space bar and another fraction appeared on the center of the screen. Fig. 2 shows a typical trial in Experiment 1.

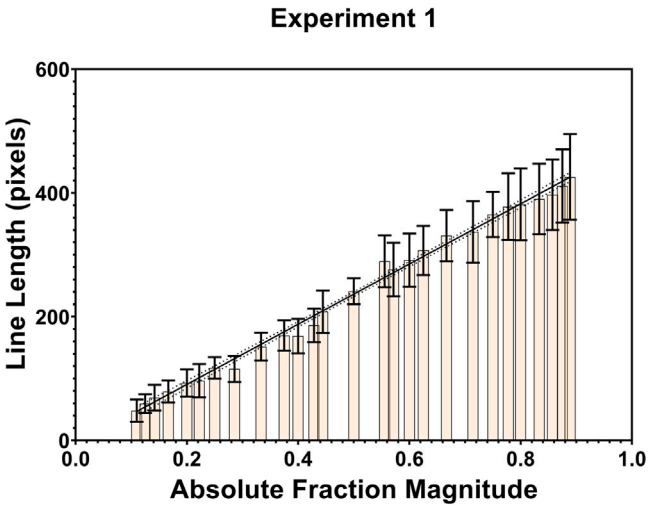


Fig. 3. Experiment 1: Mean estimated line length (in pixels) as a function of objective absolute fraction magnitude. Error bars are standard deviations of the mean.

10. Results

Fig. 3 presents the psychophysical function that related the average (across participants) lengths of the lines produced by observers in response to the objective (absolute) magnitudes of fractions. As can be noted, participants produced line lengths that corresponded very well with the objective magnitude of the fraction. We used linear regression to estimate these relations. The results captured a strong positive linear correlation between the objective and perceived fraction magnitude [$R^2 = 0.993, F(1,25) = 4065, p < 0.0001$]. Separate analyses on the data of each participants exhibited the same general pattern for 28 of the 30 participants. This may suggest that people's performance with proper fraction is comparable to that of an ideal-observer.

However, a closer look at Fig. 3 reveals that magnitude estimation was subjected to scalar variability. The standard deviation of the estimated line-length increased with the fraction's absolute magnitude. This pattern entails that estimations became noisier as the fraction's magnitude increased (e.g., fraction was closer to 1). We used linear regression to evaluate the relations between the standard deviation of an estimation and the absolute fraction magnitude [$R^2 = 0.874, F(1,25) = 174.9, p < 0.0001$], and between the standard deviation of the estimation and the estimated line length [$R^2 = 0.870, F(1,25) = 167.8, p < 0.0001$]. Fig. 4 presents the outcome of these regressions. As can be seen, the

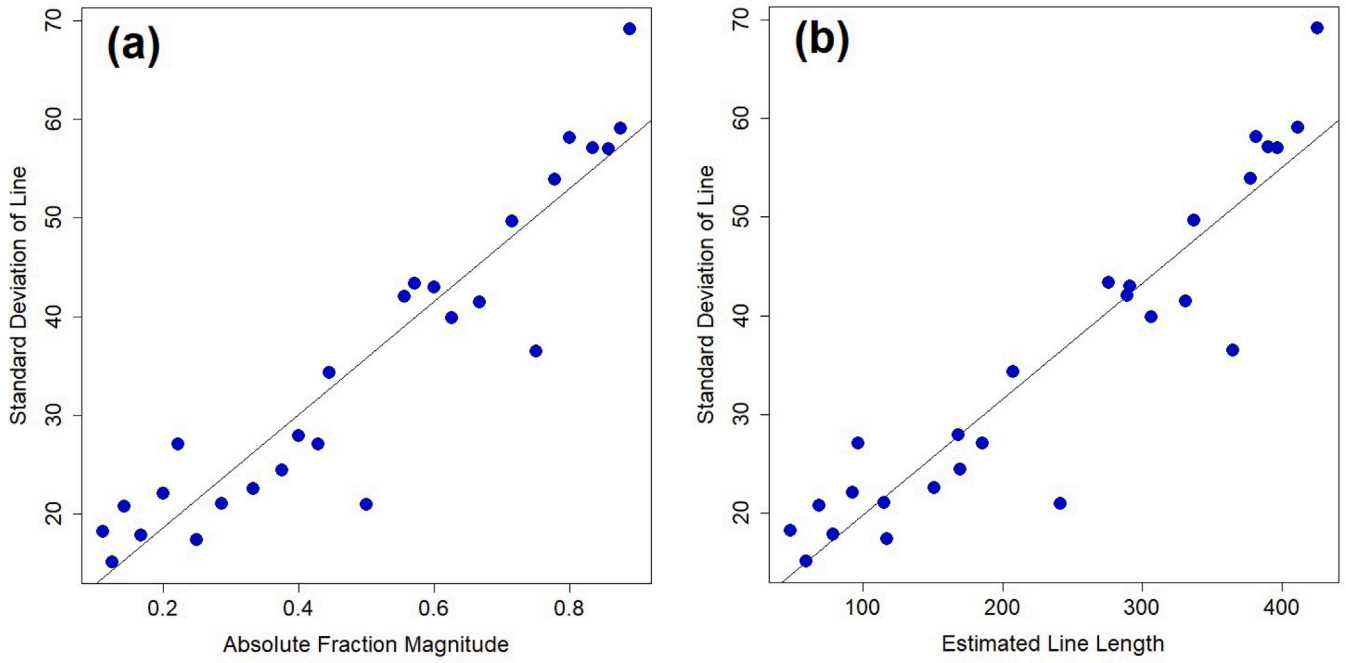


Fig. 4. Experiment 1: (a) Standard deviation of mean estimated line-length as a function of absolute fraction magnitude, and (b) Standard deviations of mean estimated line-length as a function of estimated line-length.

standard deviation was tightly correlated with the absolute magnitude of the fractions, as well as with the estimated line length. This result suggests that fractions, like whole integers are represented as a linear function that is subjected to scalar variability (Gallistel & Gelman, 2000).

Scalar variability is inconsistent with either of the strict noise-free holistic-linear or analytic-linear observer models we have simulated, because these models do not assume noisy representations. Although this finding seems most consistent with a holistic representation, it may also be accommodated by an analytic representation + motor noise. This is because the differences between the numerator and denominator become smaller as the fraction increasea in global magnitude (getting closer to 1), which requires more motor precision in producing the estimated line (see Fig. 4).

The psychophysical function depicted in Fig. 3 reflects the relations between perceived and absolute fraction magnitudes. Several of the fractions in the set convey the exact same absolute magnitude, although they are formed by different combinations of denominators and numerators (e.g. 1/2, 2/4, 3/6). Thus, this type of analysis cannot inform us on how the two components were manipulated by observers to produce the final line-length estimates. We therefore sought to further decompose the (in)dependent contributions of denominators and numerators to overall perceived magnitude. We did it first by comparing the functional measurement realization of the empirical results with the simulated results (see Fig. 1). The empirical pattern was most consistent with the holistic-linear and analytic-linear models, which are the ideal-observer models. The top panel in Fig. 5 presents this ideal-observer pattern. As can be noted, the expected line lengths are projected on the Cartesian space as a function of the denominator and numerator values, resulting in a fan shape. The fan pattern supports the idea that the overall fraction's magnitude decreases as the denominator increases, and the rate of decrease is different for each level of numerator.

The bottom panel in Fig. 5 presents the empirical results. Visual comparison supports high level of correspondences between the empirical patterns and the ideal-observer models. This becomes even more salient in the log–log plots. The correspondence was confirmed in a regression analysis with the observed estimated line length for each pair of numerator/denominator and its expected (ideal-observer) value [$R^2 = 0.992, F(1, 34) = 4482, p < 0.0001$]. The log–log plots make it clear that participants adopted an additive combination rule such that the components were treated as independent components and were added accordingly to produce the line.

11. Discussion

Participants engaged in a number-to-line task with proper fractions written in Arabic notation. Unlike previous studies, we have deployed a (semi) factorial design that consisted of all possible integers that create a proper fraction (<1), including fractions that conveyed the same magnitude, but were composed of different components (e.g. 1/2, 2/4, 3/6). This allowed us to simultaneously probe: (a) the psychophysical function that relates the absolute magnitude of the fraction to its perceived magnitude, and (b) the combination rule by which denominators and numerators were integrated. The results suggest that educated adults are highly skilled in representing proper fractions in the number-to-line task. First, the psychophysical function observed was strictly linear, entailing strong and veridical correspondence between the global magnitude of the fraction and its perceived magnitude. Second, the participants tended to adopt the optimal additive combination rule. The reader is invited to compare the ideal-observer (expected) patterns in Fig. 5A and B with their respective empirical (observed) patterns in Fig. 5C and D. However, deviations from optimality have been observed in the form of scalar variability. This finding is consistent with both holistic and analytic representation. One possibility is that the scaling of noise with global magnitude is the result of an analytic strategy. As proper fractions become larger (approaching 1), they reflect smaller distances between their components, which translate into more coarse spatial resolution (8/9 vs. 2/3). This may result in additional motor-noise component which is added to line production. Alternatively, scalar variability may reflect a holistic representation that is subjected to Weber's law. Taken together, these results are consistent with the view that educated adults can represent proper fractions in an almost optimal fashion in the number-to-line task. Second, the decomposition of the fractions reveal that participants used the noramtive division rule which is comparalbe with an additive rule (in log-log scale). Therefore, the components maintained their independence and were combined correctly. Notably, our initial simulations show that this outcome can be generated by either a holistic or analytic representation.

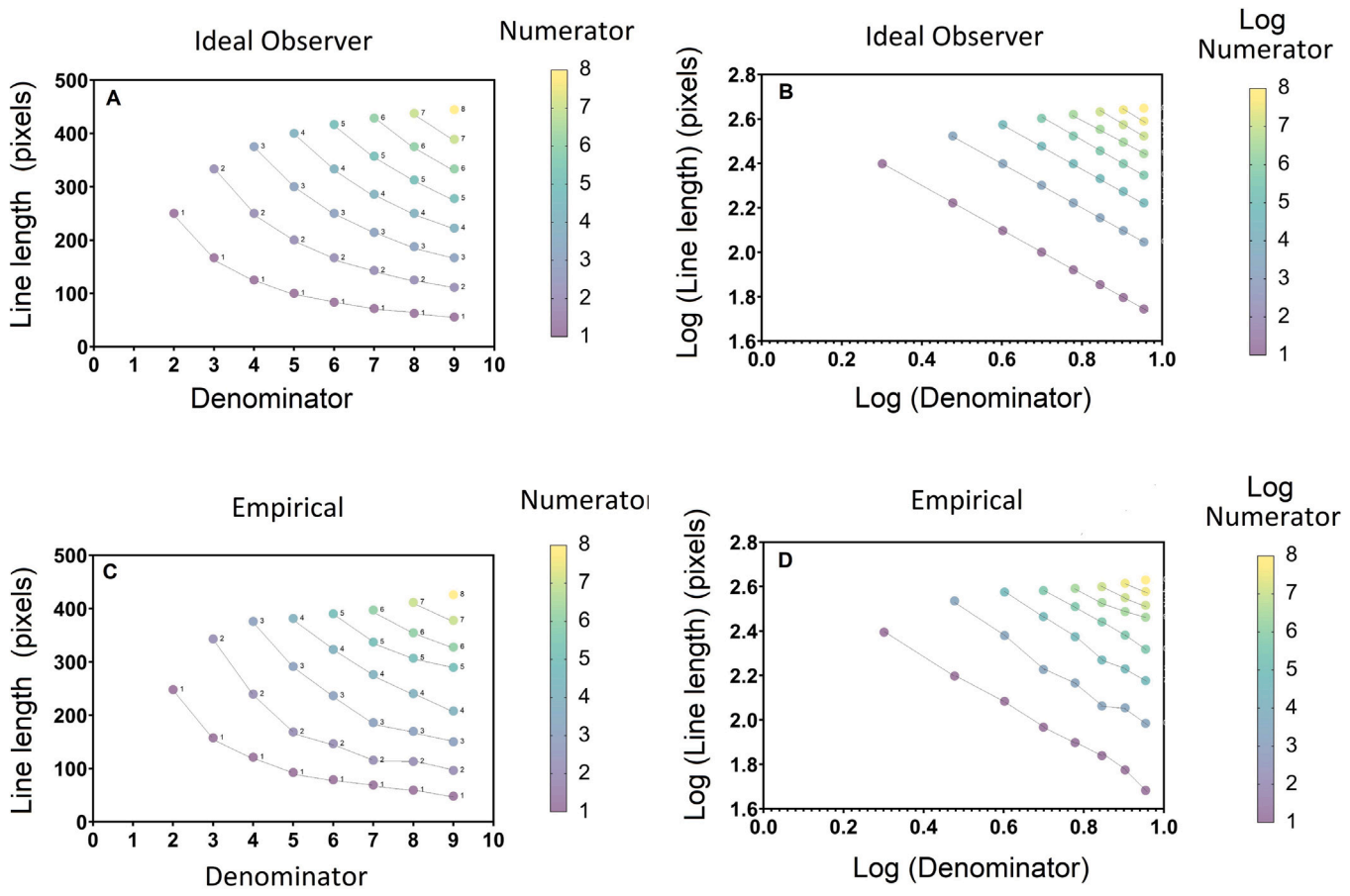


Fig. 5. Experiment 1: Functional measurement plots comparing empirical results with an ideal-observer (holistic-linear or analytic-linear): (a) Ideal observer model, (b) log–log plot of the ideal-observer model, (c) empirical results, and (d) empirical results in a log–log transform.



Fig. 6. Experiment 2: Left Panel: An example for a number-to-line trial with proper fractions written in number-word format. Right-Panel: The proper fractions in the experiment were constructed by combining each possible numerator values (1–8) with each possible denominator values (2–9). The resulting fractions in Hebrew word-notation are presented.

12. Experiment 2

The stimuli in Experiment 1 consisted of proper fractions presented in Arabic notation. But would these fractions be processed the same way had they been presented in word notation? The question of whether number processing is notation-independent has figured prominently in the numerical cognition literature (Dehaene & Akhavein, 1995). In particular, studies have shown that linguistic differences have significant impact on the processing of fractions (Miura et al., 1999; Paik & Mix, 2003). We thought that the influence of notation on the processing of fractions can provide invaluable clues into the psychological processes involved. Therefore we set to test the extent to which fractions written in word-format are processed like fractions written in Arabic notation, using the exact same techniques from Experiment 1. In the Hebrew language, the most common proper fractions are those with numerator of 1 (e.g., 1/2, 1/3, 1/4). These are denoted by single words (e.g., 1/2 is "Hetzi"). The less common fractions (the majority of proper fractions) are written as two words (e.g., 8/9 is

"Shmone Tshiyot"), which correspond almost literally to the numerator and denominator, with the word for the numerator appearing first and the word for denominator appearing second. We hypothesized that if the two formats are processed similarly (either due to a holistic-linear or analytic-linear observer model), then performance in the number-to-line task should be comparable, both with respect to the psychophysical function and with regard to the integration model. Therefore, in Experiment 2 participants performed in a number-to-line task with the exact set of proper fractions used in Experiment 1, but this time the fractions were presented in their Hebrew word notation.

13. Method

13.1. Observers

A total of 30 observers participated in this study. They were recruited from Ariel University pool of participants (Female = 27, Male = 3, mean age = 22.7, $sd = 2.87$). These participants were Psychology students of medium-to-high socioeconomic status. They were compensated with a course credit as part of an introductory course. All had normal or corrected-to-normal vision. All observers were naive to the goals of the study.

13.2. Stimuli and apparatus

The stimuli were the same fractions from Experiment 1, but this time they were written in Hebrew number words format. Fig. 6 presents the Hebrew fraction words according to their respective structure.

13.3. Procedure and design

Participants were presented with a fraction written in Hebrew words and asked to draw a line along the 0–1 scale (see Fig. 6). All the other details are identical to those in Experiment 1

14. Results

Fig. 7 presents the psychophysical function that relates objective (absolute) magnitude to perceived (produced) magnitude. As in Experiment 1, participants generated line lengths that corresponded very well with the objective magnitude of the fraction. A linear regression of the average (across-observers) line length on the absolute fraction magnitude confirmed this conclusion [$R^2 = 0.993, F(1,25) = 3493, p < 0.0001$].

One marked difference between the results of Experiment 1 and Experiment 2 is the absence of scalar variability. This conclusion was supported by a regression analysis with the standard deviation of line length estimations and the absolute fraction magnitude [$R^2 = 0.03, F(1,25) = 1.008, p = 0.325$], or the estimated line length [$R^2 = 0.04, F(1,25) = 1.079, p = 0.308$]. Fig. 8 depicts the outcome of these analyses, showing that with number-words the standard deviation was unrelated to the absolute magnitude of the fractions. This finding is important because it reveals a genuine difference in the internal representations of the two formats. Why did standard deviation scale with global magnitude for Arabic-format (Experiment 1), but not for number- word format (Experiment 2) fractions? In Experiment 1 we ascribed scalar variability to either a noisy holistic representation, or to motor noise that is added to an analytic representation. This finding may suggest that proper fractions written in different formats may be governed by two separate number systems. The word notation is governed by a precise system, while the Arabic notation invokes a more primitive and less-precise number-sense (Cordes, Gelman, Gallistel, & Whalen, 2001).

As we have already noted, the observed psychophysical function cannot fully inform us on the strategy by which participants combined

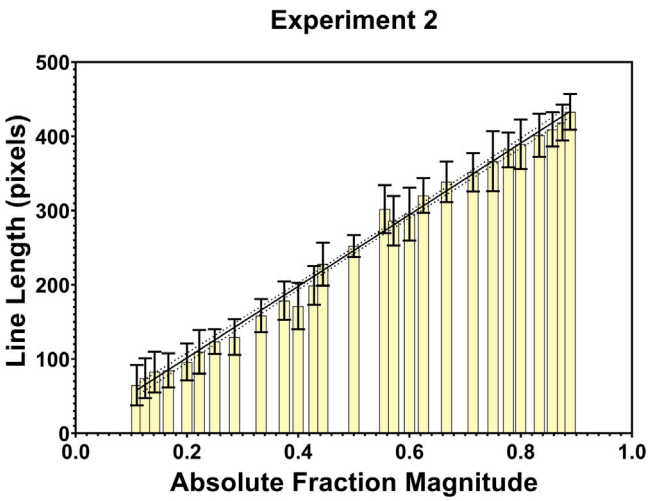


Fig. 7. Experiment 2: Mean estimated line length (in pixels) as a function of objective absolute fraction magnitude. Error bars are standard deviations of the mean.

the numerator and denominator. To uncover these strategies we decomposed the contributions of the two components on a Cartesian grid by applying the logic of functional measurement. Fig. 9 presents the empirical results of this analysis in both regular and log–log plots. Comparison of the empirical and ideal-observer (holistic-linear or analytic-linear) models (see top panel of Fig. 5) shows remarkable affinity between the two. This was also confirmed by regressing the observed estimated line length for each pair of numerator/denominator on its expected (ideal-observer) value [$R^2 = 0.992, F(1,34) = 4215, p < 0.0001$]. These results suggest that the components of fractions in word-notation are composed in an optimal fashion according to the expected additive combination rule, just like fractions written in Arabic notation. This is surprising given the difference between the two notation in scalar variability.

15. Discussion

Participants engaged in a number-to-line task with proper fractions appearing in number-word format. The psychopysical function that related the perceived magnitude to its absolute magnitude was linear. But unlike Experiment 1, standard deviation did not increase as a function of the fraction's magnitude. Estimation noise has been used in numerical cognition to dissociate between different processes in numerical cognition (Cordes et al., 2001). The nature of the variability suggests that proper fractions in word format are subjected to a different noise process than do fractions in Arabic format. One possible explanation is that fractions in Arabic notation are processed holistically and mapped onto a noisy number-line, whereas proper fractions in word notation are processed more analytically and are governed by a more accurate, algorithmic mechanism. However, in both cases we documented an additive combination rule that is consistent with ideal-observer models. It is interesting to note that in the study by Ben-Nathan and Algom (2007) with two-digit numbers, the format also did not alter the underlying combination rule.

16. Experiment 3

Experiment 1 and Experiment 2 were held with proper fractions. The current experiment set to test both proper and improper fractions. For this purpose, we presented observers with 25 fractions that were created according to a full factorial design of five denominators (i.e., 1, 3, 5, 7, and 9) and five numerators (i.e., 1, 3, 5, 7, and 9). The resulting fractions appeared in their Arabic notation (see Fig. 10). On each trial, participants were presented with a fraction, and were asked to draw a corresponding line on a 0–10 scale.

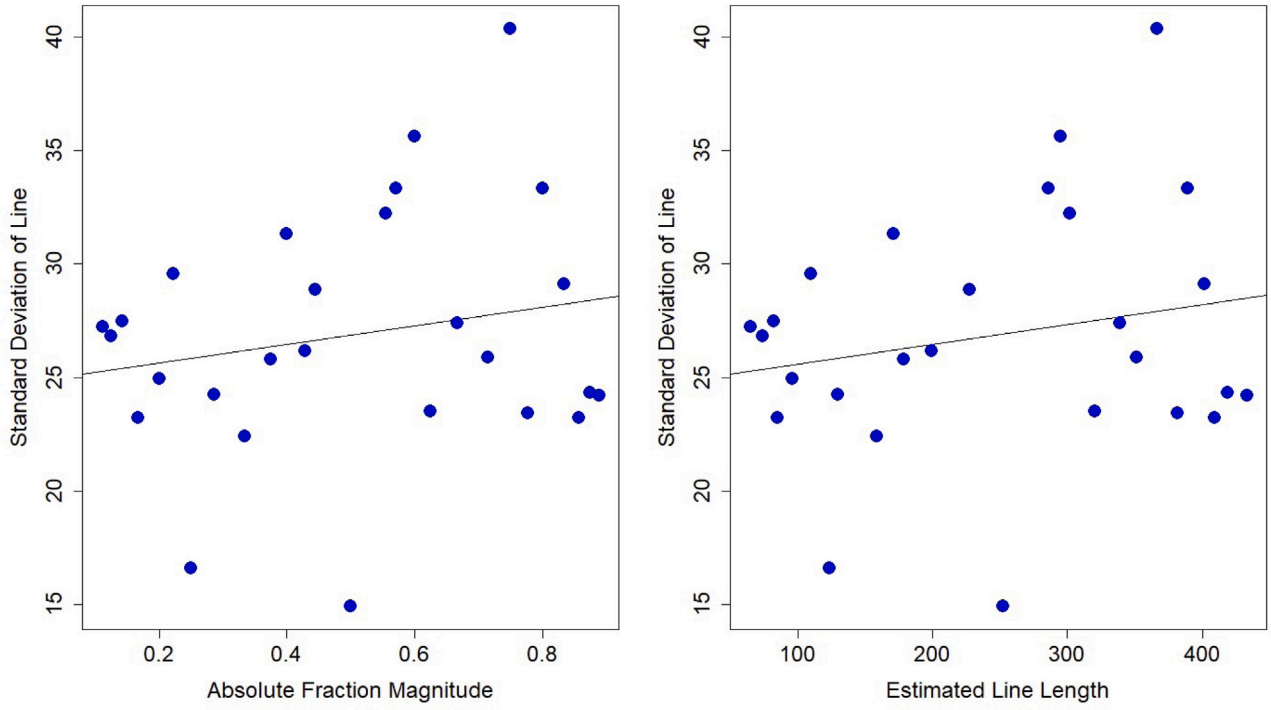


Fig. 8. Experiment 2: Left Panel: standard deviation of mean estimated line-length as a function of absolute fraction magnitude. Right Panel standard deviations of mean estimated line-length as a function of estimated line-length.

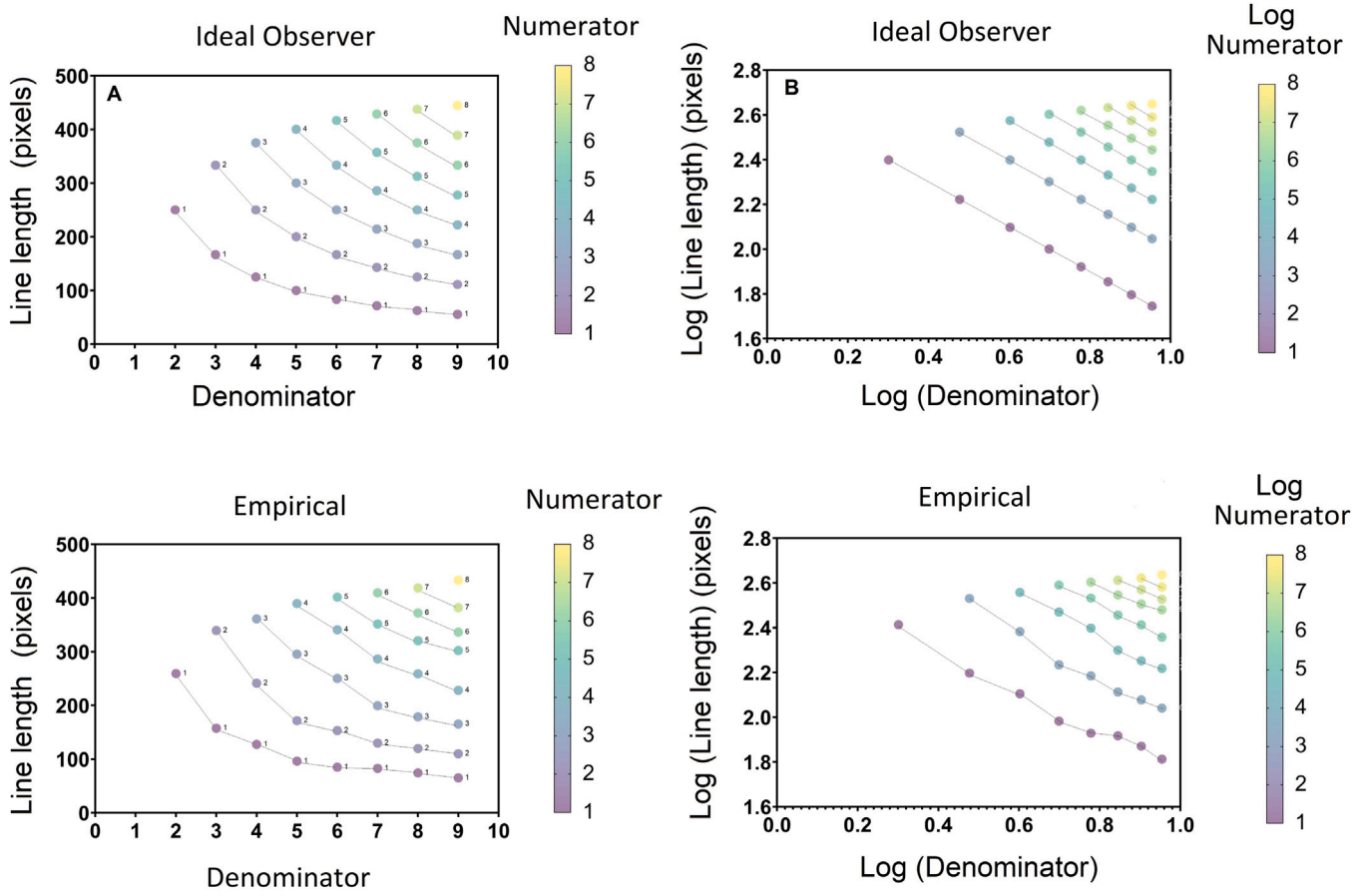


Fig. 9. Experiment 2: Functional measurement plots comparing empirical results with an ideal-observer (holistic-linear or analytic-linear): (a) Ideal observer model, (b) log–log plot of the ideal-observer model, (c) empirical results, and (d) empirical results in a log–log transform.

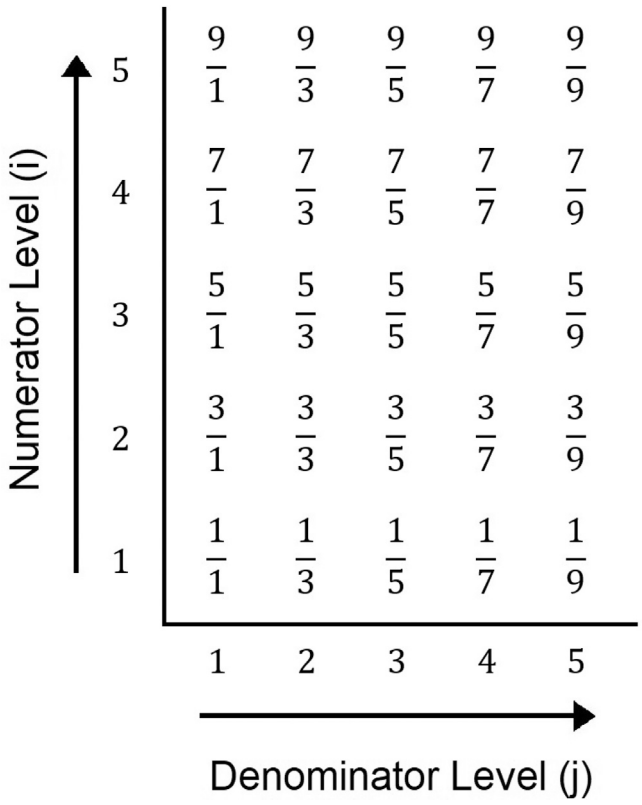


Fig. 10. Experiment 3-5: Structure of stimuli set. The fractions were constructed according to a full-factorial combination of five levels of denominator j (j = 1, 3, 5, 7 and 9) with five levels of numerator i (i = 1, 3, 5, 7, and 9). The resulting proper and improper fractions are presented.

17. Method

17.1. Observers

A total of 30 observers participated in this study. They were recruited from Ariel University pool of participants (Female = 27, Male = 3, mean age = 21.63, sd = 1.84). These participants were Psychology students of medium-to-high socioeconomic status. They were compensated with a course credit. All observers had normal or corrected-to-normal vision. All observers were naive to the goals of the study.

17.2. Stimuli and apparatus

The stimuli were composed of the numerals: 1, 3, 5, 7, and 9. These were black Arabic numerals placed over a white background. The fractions were created by a full-factorial of five levels of denominator (i.e., 1, 3, 5, 7, and 9) and five levels of numerator (i.e., 1, 3, 5, 7, and 9). Fig. 10 presents the 25 fractions on a Cartesian grid along the denominator and numerator axes. This resulted in 10 proper fractions (i.e., <1) and 15 improper fractions (i.e., ≥1). The fractions subtended the same viewing angle as reported in Experiment 1. The fractions were separated by a horizontal bar. The number-line was presented below the fraction with the numbers 0 and 10 posited at the left- and right- hand sides of it. Each experimental block consisted of 25 trials in which one of the 25 possible proper and improper fractions was chosen randomly by the computer with no replacement. Each participant completed 16 blocks, and therefore completed a total of 400 trials.

17.3. Procedure and design

These were identical to those reported in Experiment 1

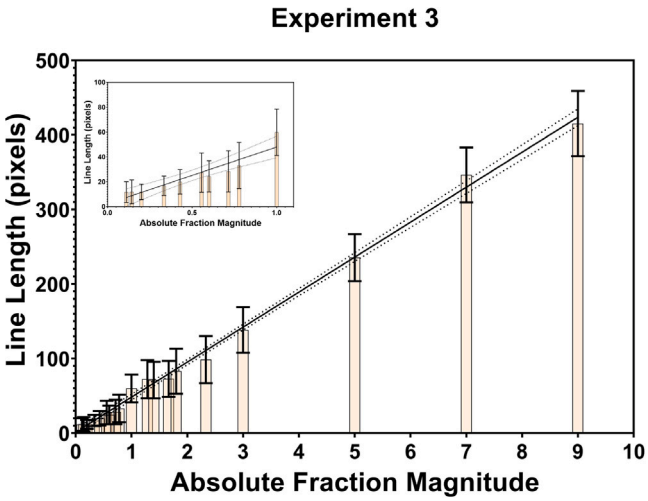


Fig. 11. Experiment 3: Mean estimated line-length as a function of objective absolute fraction magnitude. Error bars are standard deviations of the mean. The smaller graph on the top-left is zooming in on the results of the 0–1 range.

18. Results

Fig. 11 presents the psychophysial function that relates the average line length (across participants) to the global fraction magnitude. As can be noted, the function is linear [$R^2 = 0.99, F(1,17) = 65.26, p < 0.0001$], capturing the same pattern as in the two previous experiments which presented only proper fractions.

Recall that in Experiment 1, we found evidence for scalar variability. Here, we tested whether such a trend emerges when a mixed-set of improper and proper fractions is used. For this end, we plotted the standard deviation of line-length estimations as a function of the fraction's absolute magnitude (see Fig. 12). As can be noted, the two quantities exhibited lawful relations. The estimated line became more noisy as the absolute magnitude of the fraction increased. A linear regression of the standard deviation on the absolute fraction magnitude [$R^2 = 0.74, F(1,17) = 7.03, p < 0.001$], and on the estimated line length [$R^2 = 0.73, F(1,17) = 6.79, p < 0.001$] supported this observation. However, a closer look at Fig. 12 suggests that, in contrast to Experiment 1, variability did not increase in a linear fashion, but rather obeyed a diminishing returns function. This pattern can be interpreted in two possible ways: (a) the points obey a two-limb function, with a steep linear function for proper fractions, and a moderate linear function for improper fractions, (b) the points reflect a logarithmic or power function with an exponent smaller than 1. Given the outcome of Experiment 1, possiblity (a) is more plaubsible than (b). In any event, this pattern of noise variability suggests that different processing mechanisms are involved in the perception of fractions in a mixed-set of fractions compared to proper-only set of fractions.

Next, we sought to uncover the strategy by which participants combined the numerator and denominator of fractions in this mixed-set. We therefore decomposed their separate contributions to the produced line length according to the logic of functional measurement. Fig. 13 presents the results. Eyeballing the empirical plots reveals a slight deviation from additivity in the log–log sclae. The full factorial design of this experiment allowed us to conduct a two-way ANOVA with 5 levels of numerator and 5 levels of denominator, with the logarithm of the produced lines serving as a dependent variable. As we recounted earlier, the logarithmic transformation predicts a family of linear decreasing functions that reflect an additive combination rule. Any deviation from additivity in this plot should be captured by the ANOVA as a statistically significant interaction (Anderson, 1981). In this analysis we found a main effect of numerator value [F(1,742) =

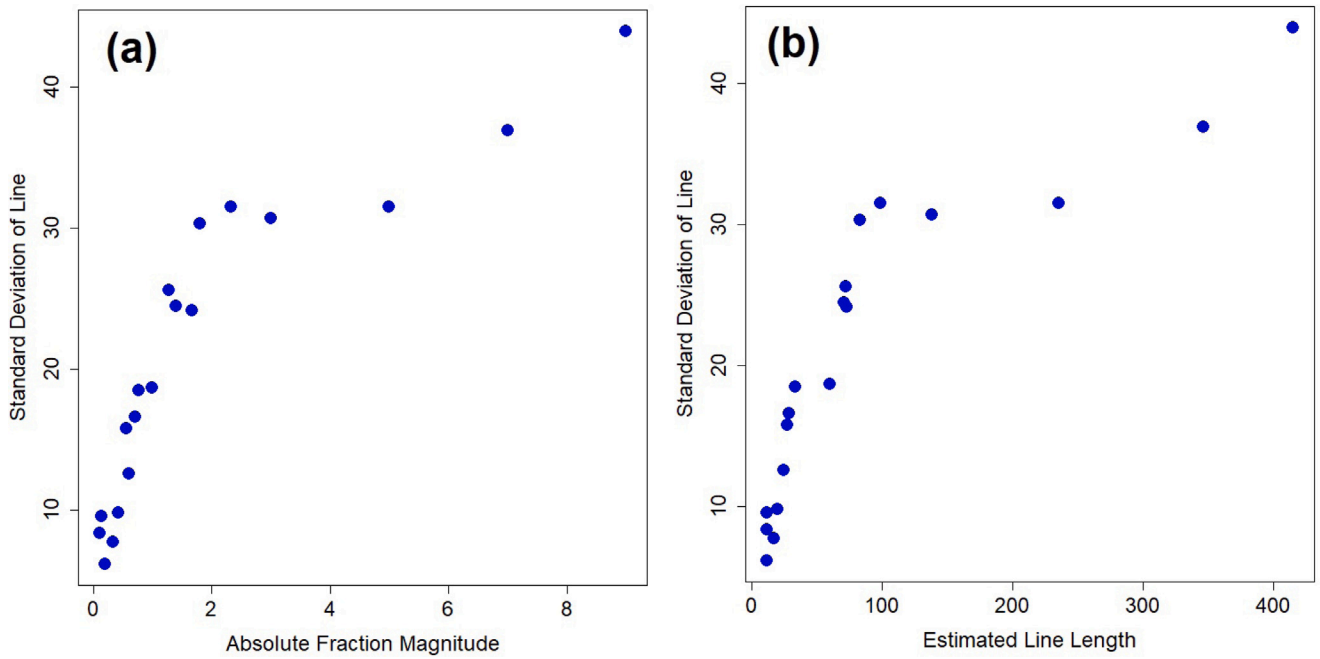


Fig. 12. Experiment 3: (a) Standard deviation of mean perceived line-length as a function of absolute fraction magnitude, and (b) Standard deviation of mean perceived line-length as a function of estimated line-length.

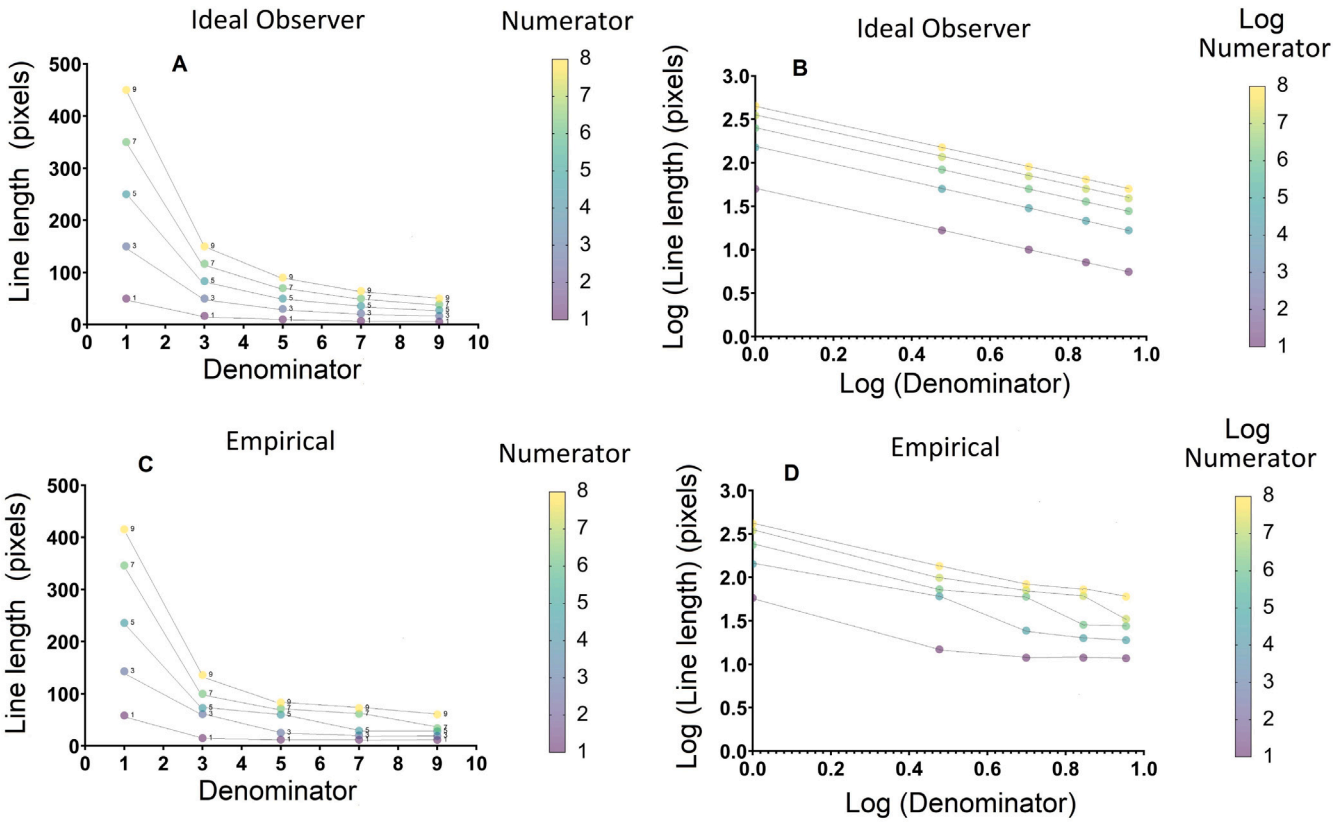


Fig. 13. Experiment 3: Functional measurement plots comparing empirical results with an ideal-observer (holistic-linear or analytic-linear): (a) Ideal observer model, (b) log–log plot of the ideal-observer model, (c) empirical results, and (d) empirical results in a log–log transform.

316.51, MSE = 19.78, p < 0.0001], a main effect of denominator value [F(1,742) = 319.95, MSE = 19.99, p < 0.0001], but no interaction [F < 1]. This result suggest that the numerator and denominator were optimally integrated according to an additive rule.

19. Discussion

Participants performed in a number-to-line task with a mixed set of proper and improper fractions. The structure of the stimuli set followed a full factorial with five levels of denominator and five levels of numerator. This allowed us to assess both the psychophysical function and the decomposition rule that guided the perception of these fractions. The results were similar to those observed in Experiment 1. First, objective and subjective magnitudes corresponded very well according to linear function. We found that the estimation got noisier as the overall magnitude of the fraction increased. Second, participants' performance was compatible with an additive optimal integration rule.

20. Interim summary Experiments 1–3

Experiments 1–3 were set to test the internal representations that govern the processing of symbolic fractions. Participants engaged in a number-to-line task with proper fractions written in Arabic or word formats (Experiments 1–2) and with a mixed-set of proper and improper fractions written in Arabic notation (Experiment 3). The results were generally consistent across all three experiments, suggesting that educated adults can represent fractions quite accurately by integrating denominators and numerators according to an optimal additive integration rule. With Arabic notation, the subjective estimation got noisier as a function of the fraction's global magnitude. These results require additional converging operations. We note that the internal representation underlying numerical information might change as a function of the task adopted (Fitousi & Algom, 2020), or that educated adults acquire and use, not one but several types of representations (Siegler et al., 2009). In the next two experiments, we have used a none-speeded comparative judgment task with the same factorial set of fractions as in Experiment 3. We modeled the data with the MLCM — maximum likelihood conjoint measurement technique (Knoblauch & Maloney, 2012; Luce & Tukey, 1964), which allowed us to test various models of integration, including the optimal additive strategy.

It should be noted that strategies used for mapping fractions onto a number line may vary both across fractions and across participants. For instance, behavioral performance for benchmark fractions (e.g., 1/2, 3/4) may differ significantly from non-benchmark fractions; Some fractions may be more easily transformed into decimals than other fractions, and some participants may tend to use the "decimal transformation" strategy more often than others. A closer look at the functional measurement fan in Figs. 5, 9, and 13 reveals that benchmark fractions were subjected to the same transformation as other fractions. Eliminating these benchmark fractions will evidently result in the same psychophysical function.

Another point concerns the limitation of our stimuli, which included only fractions in the 0–1 or 0–10 range. It has been shown that linear estimation for whole numbers on number line emerges later for larger scale (e.g., 0–1000) than smaller scale (e.g., 0–100) number line (Booth & Siegler, 2006). It is possible that the magnitude of numerators and denominators impact how linear the fraction magnitude is represented (e.g., consider 7/82 where 7 could be linearly represented but 82 could be logarithmically represented). This is obviously a possibility that should be tested in future research with the tools applied here.

21. Experiments 4–5: Maximum likelihood conjoint measurement

A fraction is a multidimensional object that is composed of two primary dimensions — a numerator and a denominator. To compare two symbolic fractions, an ideal-observer should first extract the global magnitude of each fraction by integrating the information from the two dimensions, then compare these magnitudes, and finally choose the larger/smaller value. Application of the Maximum Likelihood Conjoint Measurement (MLCM, Knoblauch & Maloney, 2012; Luce & Tukey, 1964) allowed us to assess human performance against such an ideal-observer. In particular, MLCM afforded us to *simultaneously* measure and model the contributions of both numerator and denominator to perceived global magnitude. The results are estimates of two functions: *denominator* and *numerator* magnitudes that capture these contributions. The task requires a comparative judgment of pairs of fractions. The stimuli are combined from five levels of numerator magnitudes and five levels of denominator magnitudes (Knoblauch & Maloney, 2012).

Participants viewed all possible 325 pairs of the 25 fractions (see Fig. 10), and on each trial judged which member of the pair conveyed a larger numerical quantity. Let ϕ^N denote values on the numerator magnitude scale and let ϕ^D denote values on the denominator magnitude scale. As can be seen in Fig. 10, the value of numerator is constant across the ith row, and is denoted ϕ_i^N and the value of denominator is similarly constant in the jth column and is denoted ϕ_j^D. Comparison between two fractions can be represented as a comparison between two ordered pairs (ϕ_i^N, ϕ_j^D) and (ϕ_k^N, ϕ_l^D). It is assumed that participants compute an estimate of perceived fraction magnitude based on both its numerator ϕ^N and denominator ϕ^D. The estimates of this quantity are $\psi(\phi_k^N, \phi_l^D)$ and $\psi(\phi_i^N, \phi_j^D)$, for the first (F1) and second (F2) fraction respectively. These can also be written as $\psi(\phi_1^N, \phi_1^D)$ and $(\psi(\phi_2^N, \phi_2^D))$, respectively. The resulting estimates of perceived fraction magnitude amounts to:

$$\Delta(F1, F2) = \psi_1 - \psi_2 + \varepsilon = \psi(\phi_1^N, \phi_1^D) - \psi(\phi_2^N, \phi_2^D) + \varepsilon = \delta + \varepsilon \quad (5)$$

where ψ_1 and ψ_2 are internal representations for the magnitude of the first and second fractions, respectively, determined by the psychophysical function, ψ. ε is a Gaussian random variable with mean $\mu = 0$ and variance σ^2 representing judgment error and Δ is the decision variable. It is assumed that the observer chooses the first fraction when $\Delta > 0$, and otherwise the second. The log-likelihood of the model over all trials given the observer's response is:

$$LL(\Delta_i, R_i) = \sum_i R_i \log(\phi(\delta/\sigma)) + (1 - R_i)\log(1 - \phi(\delta/\sigma)) \quad (6)$$

where R_i is the response in the ith trial. It can take the values of either 0 or 1 (for choosing either the first or the second fraction of the pair), and ϕ is the cumulative distribution function for a standard normal variable. We describe three models to which we estimated the *maximized likelihood* of the psychological responses ψ's across all trials. Specific constrained are also described that render the models identifiable (Knoblauch & Maloney, 2012). There are at least three nested models of the decision process that can be tested in the MLCM framework (Ho et al., 2008; Knoblauch & Maloney, 2012): (a) *the independent observer*, (b) *the additive observer* (which is an ideal observer in our case), and (c) *the saturated observer*. Under the independence model the observer bases their decision on the denominator (numerator) and the decision variable is:

$$\Delta = \psi_1^D - \psi_2^D + \varepsilon \quad (7)$$

where ψ_1^D and ψ_2^D are the internal representations of fraction magnitude evoked by the values of the denominators of fraction 1 and 2, respectively. A comparable model is defined to model the independent contributions of the values of numerators in the two fractions with N's replacing D's.

In the additive model the decision variable is the sum of the numerator and denominator signals

$$\Delta = (\psi_1^D - \psi_2^D) + (\psi_1^N - \psi_2^N) + \varepsilon \quad (8)$$

where the denominator and numerator terms are regrouped to demonstrate that the observer is comparing perceptual intervals along one dimension with perceptual intervals along the other.

In the third interaction (saturated) model, non-additive combination terms are introduced. The decision variable is written as:

$$\Delta = (\psi_1^D - \psi_2^D) + (\psi_1^N - \psi_2^N) + (\psi_1^{DN} - \psi_2^{DN}) + \varepsilon \quad (9)$$

where the rightmost term corresponds to an interaction that depends on each numerator–denominator combination.

21.1. Ideal-observer

We assume that whenever an ideal-observer is presented with a pair of fractions for comparative judgment he or she acts according to the following: (a) extracts the individual magnitude of each component, (b) divide the numerator by the denominator for each fraction, (c) compares the outcomes of the two operations, and (d) indicates the larger of the two. In case of parity (3/3 5/5), the observer gives a random response. Note that treating the denominator and numerator as independent entities or as interactive entities (e.g., due to skewed number line for example) is sub-optimal. An optimal strategy is that of following the division operation closely. Recall, that according to a fundamental axiom in *foundation of measurement* (Krantz et al., 1971; Luce & Tukey, 1964), conjoint psychological representations are consistent with any additive operation (e.g., addition, subtraction). This implies that in the case of division, an additive model is the optimal strategy. We simulated data that reflected the performance of such an ideal-observer and found indeed that the additive model provided the best fit to the simulated data. Fig. 14 presents our MLCM analyses on these simulated data. At the top panel of Fig. 14 we illustrate the perceptual scale values as a function of the objective scale values, separately for the denominator and numerator. These estimates can inform us on how an ideal-observer transforms the values of the numerator and denominator onto a perceptual (psychological) scale of magnitude. As can be noted, the perceived magnitude of a fraction decreases as the magnitude of the denominator increases. In contrast, the perceived magnitude of a fraction increases as the numerator increases.

When the denominator and numerator values are combined according to the correct normative (division) rule, then MLCM predicts that the psychological contributions should obey an additive integration observer model. Such an ideal-observer pattern can be seen at the bottom-panel of Fig. 14. The Figure presents the estimated perceived global fraction magnitude as a function of the denominator values, at each level of numerator value. The pattern bears close affinity with the ideal-observer plot for the functional measurement analysis of Experiment 3 (compare to Panel B in Fig. 13). However, the MLCM plot is derived from ordinal judgments, whereas the conjoint measurement plot is derived from interval scale judgments. Any deviation from this pattern, either due to a saturated observer model or to an independent observer model would entail sub-optimal integration. An additive observer model is consistent with either holistic or analytic representations (see Fitousi, 2022). It is tempting to suggest that a saturated model is likely the result of a holistic magnitude representation, but analytic representations can also produce this observer model. The least expected model is the independent observer model, which is consistent with a decomposed representation. It is not expected because according to this model the two components are not combined at all. Practically, the independent model implies a complete lack of knowledge of fractions, and would result in a high rate of errors.

22. Experiment 4

The goal of this experiment was to test the integration process by which denominator and numerator are combined to produce the perceived fraction's global magnitude. Participants performed in a non-speeded comparative judgment task with the same fractions from Experiment 3. The members of a pair were presented sequentially. We assumed that sequential mode of presentation enhances holistic representation compared to parallel mode of presentation (see Experiment 5). MLCM models will be fit to the empirical data and compared to ideal-observer performance.

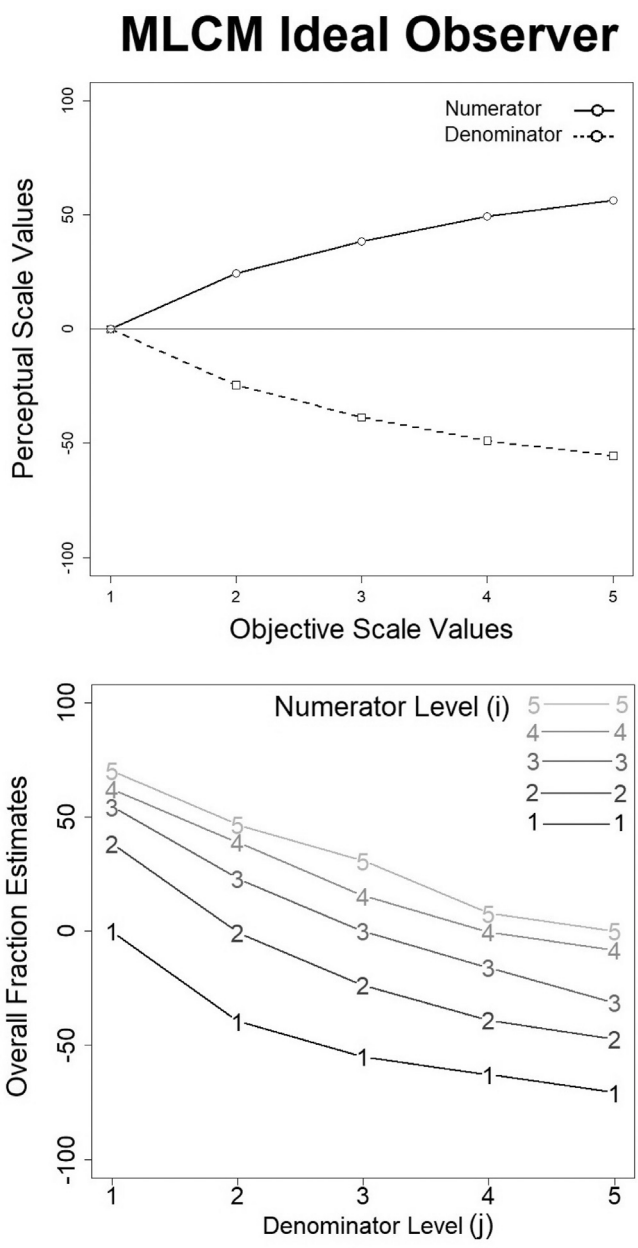


Fig. 14. Experiments 4–5: Ideal-observer analyses according to MLCM: Top-Panel: Estimated perceptual scale values as a function of objective scale values, separately for numerator and denominator. Bottom-Panel: Overall fraction estimates as a function of denominator level for each level of numerator. Note that the numbers 1–5 in this Figure stand for the *i* numerator level and are not the actual numerator values (1 = 1, 2 = 3, 3 = 5, 4 = 7, and 5 = 9).

23. Method

23.1. Participants

A total of 20 observers participated in this study. They were recruited from Ariel University pool of participants (F = 18, M = 2, mean age = 22.7, *sd* = 1.71). Observers were compensated with a course credit. All observers had normal or corrected-to-normal vision. All observers were naive to the goals of the study. The MLCM is a psychophysical technique and its analyses are held at the level of the individual (Luce & Tukey, 1964). The choice of sample size was guided by previous MLCM studies (Ho et al., 2008).

23.2. Stimuli and apparatus

The same set of stimuli from Experiment 3 was used. These were created with the Microsoft Paint software. All numerical characters were in font type 'Ariel'. Five levels of numerator (1, 3, 5, 7, and 9) and five levels of denominator (1, 3, 5, 7, and 9) were used. The five numerator and five denominator levels were combined in a full factorial fashion to yield 25 possible fractions (5 numerator levels × 5 denominator levels = 25). The fractions were presented in a canonical view with a horizontal line separating the two components. Fig. 10 presents the stimuli on a Cartesian grid. Let i denote the rows of this grid and j denote its columns. The stimuli matrix represents a full factorial combination of i levels of numerator magnitudes and j levels of denominator magnitudes. For a given level of denominator j, numerator is increasing as we move up the ith rows. For a given numerator level i, the denominator magnitude is increasing as we move to the right jth columns.

23.3. Procedure and design

On each trial observers viewed 1 of the 325 possible pairs of fractions from those illustrated in Fig. 10. Viewing distance was approximately 56 cm. The two fractions were presented one after the other. A fixation point appeared for 200 ms, then the first fraction was presented for 600 ms and removed from the screen; then the second fraction was presented for another 600 ms. This type of presentation was used to facilitate holistic processing and to enable comparison with a parallel mode of presentation (tested in Experiment 5). The observer's task was to judge which of the two fractions was numerically larger. Each pair was presented 3 times, making a total of 975 trials.[4] The observer indicated whether the first fraction or the second fraction was numerically larger by pressing one of two corresponding keys on the keyboard ('z' or 'm'). The observer's response initiated the next trial. The top panel of Fig. 15 illustrates a typical trial sequence in Experiment 4.

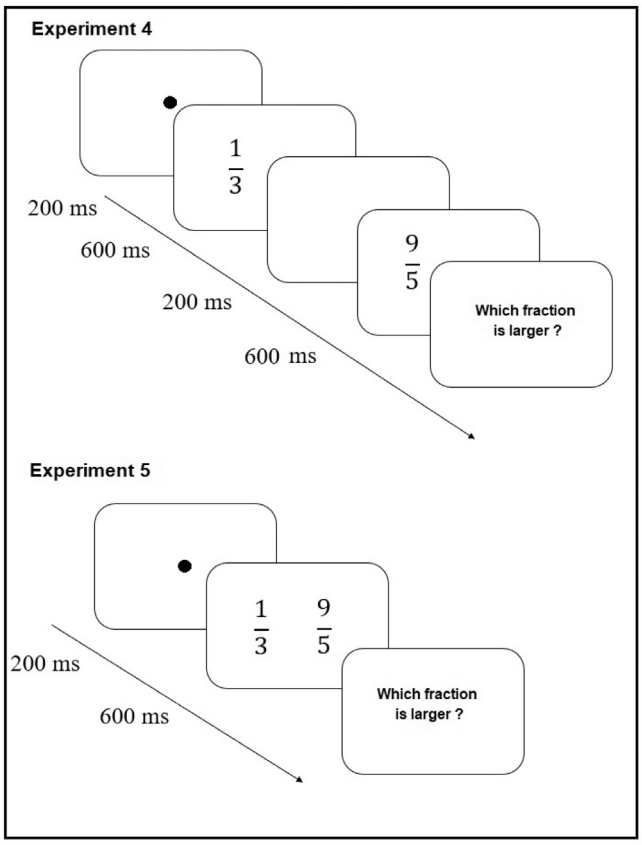


Fig. 15. Experiment 4–5: Trial dynamic in Experiment 4 (Top Panel) and Experiment 5 (Bottom Panel).

24. Results

24.1. Accuracy

The performance of two participants (4 and 19) was below chance level and they were removed from analyses. This is not surprising given adults' general difficulties with fractions. Overall accuracy of the 18 remaining participants amounted to 84% correct ($sd = 3.5\%$). We next computed the absolute distance between every pair of fractions and tested for the presence of a distance effect by performing a linear regression with the log of accuracy and the log of absolute distance. The results [$R^2 = 0.38, F(1, 123) = 75.67, p < 0.0001$] confirmed the presence of a global distance effect. As can be noted in Fig. 16 overall accuracy increased as a function of global distance, and this finding held regardless of whether the participant was shown two proper fractions, two improper fractions, or a combination of the two. The finding of a distance effect with fractions (DeWolf et al., 2014; Schneider & Siegler, 2010), as with other two-dimensional mathematical objects, such as two-digit numbers (Fitousi & Algom, 2020), has been often taken to support holistic processing.

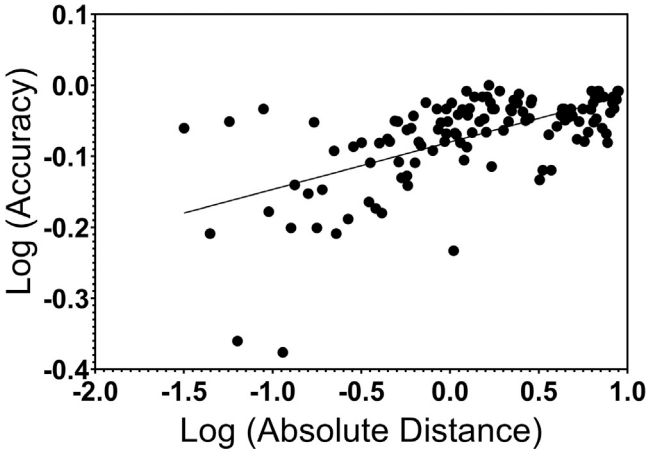


Fig. 16. Experiment 4: A distance effect for global fraction magnitude. Log Accuracy as a function of Log absolute distance.

24.2. MLCM

Data analyses were performed with the MLCM package (Knoblauch, Maloney, & Aguilar, 2014) in the open source software R (R Core Team, 2017). The default method uses the GLM function to estimate the model coefficients with a maximum likelihood criterion. The ML estimates are found with an iteratively reweighted least squares (IRLS) algorithm. The χ^2 statistic is computed from the differences of deviance of the nested model fits (see Knoblauch & Maloney, 2012, pp. 240–245).

We first compared the additive (ideal-observer) and independent models separately for each observer. The degrees of freedom for this specific test were computed as the difference between the coefficient

[4] This is the smallest number of repetitions needed to produce reliable parameter estimates (Ho et al., 2008).

Table 1
Experiment 4: Comparison of independent and additive observer models for judgments of fraction magnitude. The Bonferroni-corrected level for multiple comparisons is p < .0027. Two participants (4 and 19) exhibited accuracy rates below chance and were not included.

	Test fractions (additive model x independent model)		
	DF	χ^2	*p*-value
Obs. 1	4	385.49	<.001***
Obs. 2	4	530.12	<.001***
Obs. 3	4	212.44	<.001***
Obs. 5	4	181.66	<.001***
Obs. 6	4	346.80	<.001***
Obs. 7	4	251.87	<.001***
Obs. 8	4	351.59	<.001***
Obs. 9	4	387.15	<.001***
Obs. 10	4	506.34	<.001***
Obs. 11	4	544.55	<.001***
Obs. 12	4	639.91	<.001***
Obs. 13	4	112.54	<.001***
Obs. 14	4	193.45	<.001***
Obs. 15	4	287.02	<.001***
Obs. 16	4	385.62	<.001***
Obs. 17	4	363.07	<.001***
Obs. 18	4	419.74	<.001***
Obs. 20	4	87.67	<.001***

*** p < 0.0027.

Table 2
Experiment 4: Comparison of additive and saturated observer models. The Bonferroni-corrected level for multiple comparisons is p < .0027. Two participants (4 and 19) exhibited accuracy rates below chance and were not included.

	Test fractions (saturated model x additive model)		
	DF	χ^2	*p*-value
Obs. 1	16	24.03	0.08
Obs. 2	16	45.21	0.001***
Obs. 3	16	19.48	0.24
Obs. 5	16	51.89	0.001***
Obs. 6	16	47.63	0.001***
Obs. 7	16	34.73	0.001***
Obs. 8	16	66.45	0.001***
Obs. 9	16	40.05	0.001***
Obs. 10	16	46.88	0.001***
Obs. 11	16	18.32	0.30
Obs. 12	16	26.92	0.04
Obs. 13	16	28.14	0.03
Obs. 14	16	20.57	0.19
Obs. 15	16	22.49	0.12
Obs. 16	16	40.01	0.001***
Obs. 17	16	57.53	0.001***
Obs. 18	16	25.23	0.06
Obs. 20	16	17.10	0.37

*** p < 0.0027.

estimates in each model (8 for the additive model - 4 for the independence model = 4). The additive model exhibited better fit to the data than the independent model for all participants (see Table 1). This is expected given that a completely independent model entails a complete lack of knowledge of symbolic fractions. Such a model produces very high error rates and is clearly untenable.

We next derived the estimated contributions of denominator and numerator values to the perceived global fraction magnitude. These estimates can inform us on how observers transform the values of denominators and numerators onto a perceptual (psychological) scale of global fraction magnitude. Panel A in Fig. 17 plots the predictions for an ideal-observer. The plot for the actual data is presented in Panel C in Fig. 17. The two are quite comparable. The perceived magnitude of a fraction monotonically increases as its numerator increases; but it decreases monotonically as its denominator increases. Panel B plots the data of participant 11 whose performance conformed with an ideal-observer pattern, whereas Panel D presents the data of participant 8, whose performance did not obey the ideal-observer pattern. However, a more critical test compares the saturated and the additive models (see Table 2). The χ^2 statistics in this case has 16 degrees of freedom because it is computed as the difference in number of free parameters of the less constrained model (the saturated model = 24) and the more constrained model (the additive model = 8). At a Bonferroni-corrected level of $p < .0027$, the saturated model exhibited better fit for half (9/18) of the participants (see Table 2). This outcome suggests that half of the participants combined fractions' components according to a saturated (suboptimal) observer model.

Panel A in Fig. 18 presents the expected pattern for an ideal-observer. Panel B in Fig. 18 presents the pattern of participant 11, whose data conformed to that of an ideal-observer. As can be noted, the two are quite comparable. In both, one can note a family of parallel decreasing functions. However, Panel D in 18 illustrates the pattern of participant 8, whose data was best fit by a saturated observer model, and therefore did not conform to an ideal-observer pattern. As can be noted, the lines are not parallel, and as such, reflect an interactive pattern. This pattern signals deviation from additivity, and therefore a suboptimal integration strategy. Note that the ideal-observer is an additive model by which numerators and denominators are combined according to the normative division rule. This entails that global fraction magnitude should be decreasing as the denominator is increasing, and the rate of the decrease should be held constant across levels of numerator (see Fig. 18a). For example, Observer 11 (see Fig. 18b) closely mimicked this pattern. However, when a participant exhibited significant deviations from the additive model, the resulting estimation of the compound fraction did not conform to the correct normative model, resulting in an interactive fan pattern. Such observers do not apply the division rule correctly or do not represent the components accurately. This can occur due to biases in the representation of the individual components or because the participant has not fully acquired the notion of division. Take for example, Observer 8 (see Fig. 18c), who has exhibited significant deviations from the additive pattern. For this participant, the psychological magnitude of the fraction 7/9 appears to be larger than that of the fraction 7/7. Another strong deviation in the data of this participant can be seen in their estimation of fractions with a numerator of 9. The expected decline in estimated magnitude of these fractions (9/1, 9/3, 9/5, and 9/9) as a function of denominator is violated, such that the psychological magnitude of 9/5 is larger than that of 9/3. What type of psychological mechanisms can generate such biases? One possible mechanism is the need to resolve the conflict between incongruent sources of information — the magnitude of the components on the one hand, and the magnitude of the fraction, on the other. Thus, some participants may be misled by the sheer magnitude of the components to the effect that fractions with larger components are perceived as larger than what they are. This phenomenon bears some resemblance to a *Stroop effect* (Fitousi & Algom, 2006). Another possible mechanism is *anchoring* (Tversky & Kahneman, 1974). According to this idea, some participants may use the magnitude of the components as an anchor for the overall fraction magnitude. When the components (numerator and/or denominator) are large they will use a higher anchor, which results in overestimation of the fraction's magnitude.

Another tool that can help us in testing for the presence/absence of optimal integration is the response proportion plot (Ho et al., 2008; Knoblauch & Maloney, 2012) which is provided by the MLCM package (Knoblauch et al., 2014). Panel A in 19 shows the proportions of responses in the simulated data of an ideal-observer. The reader can compare this pattern with that of participant 11 (Panel B), whose data were consisted with an optimal (additive) integration; and with that of participant 8 (Panel D) whose data were inconsistent with an optimal integration.

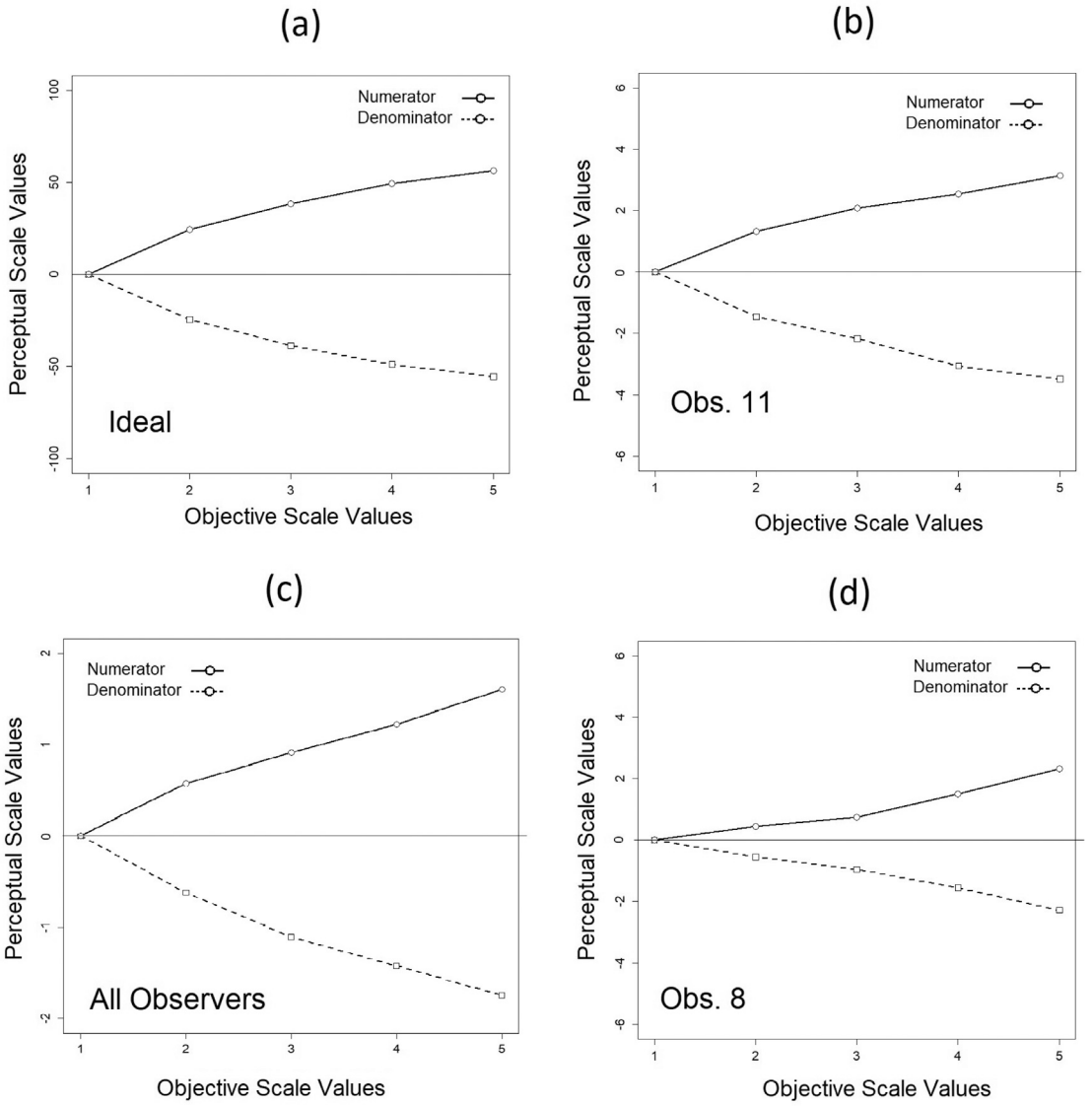


Fig. 17. Experiment 4: Estimated perceptual scales as a function of objective values obtained by fitting the additive MLCM model. Panel (a): Ideal Observer (additive model) pattern, (b) Participant 11 whose data conformed with an ideal observer, (c) all observers, and (d) Participant 8 whose data violated the ideal observer.

25. Discussion

In this experiment we applied the MLMC (Knoblauch et al., 2014) to test whether educated adults combine fractions' components in an optimal fashion. Participants performed in a comparative judgment task with proper and improper fractions that were constructed according to a factorial design. The fractions were presented sequentially. First, we found a global distance effect on accuracy, an effect that is consistent with holistic processing. Second, half of the participants exhibited the (ideal-observer) additive model of integration, whereas half of participants integrated the components according to a saturated observer model. This latter model is sub-optimal because it is deviates from the correct normative mathematical rule. These results might come as a surprise given the optimal performance we found in the number-to-line tasks in Experiments1-3. However, one should recall that the comparative judgment task is more difficult than the number-to-line task. First, it involves two fractions rather than one. Second, unlike the number-to-line task, it does not rely on a visually transparent medium such as spatial location. Third, it requires more sophisticated mechanisms than estimation because it requires to compare the actual magnitudes of two fractions. This may lead participants to adopt different types of representations in the two tasks. However, the fractions in the current experiment were presented sequentially, and this mode of presentation may also have affected the way participants represented the magnitude of fractions. Thus in Experiment 5 we presented the fractions simultaneously rather than sequentially.

26. Experiment 5

This experiment was similar to Experiment 4, with the only exception that the fractions were presented simultaneously rather than sequentially

27. Method

27.1. Participants

A new group of 24 observers from the Psychology department who did not participate in Experiment 3 took part in this experiment. They were recruited from Ariel University pool of participants (F = 21, M = 3, Mean age = 22.7, sd = 2.47). They were compensated with a course credit.

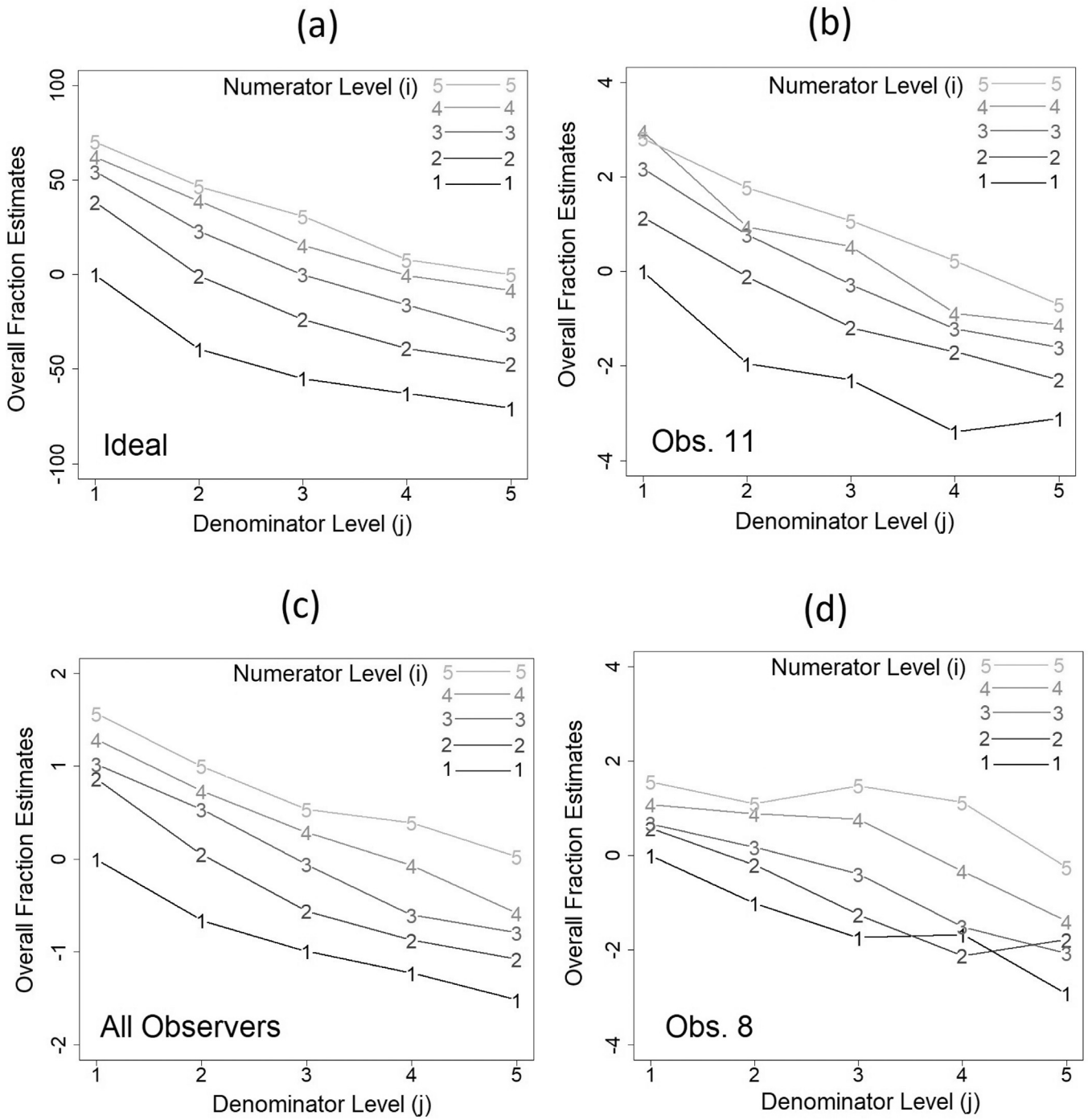


Fig. 18. Experiment 4: Overall fraction estimates as a function of denominator level for each level of numerator. Panel (a): Ideal-Observer (additive model), (b) Empirical results for participant 11 who conformed with the ideal-observer (additive) model, (c) Empirical results across all-participants, and (d) Empirical results for participant 8 who whose data were most consistent with a saturated observer model, which violates the ideal-observer (additive) model. Note that the numbers 1–5 in this Figure stand for the *i* numerator or *j* denominator levels and are not the actual numerator or denominator values (the mapping is 1 = 1, 2 = 3, 3 = 5, 4 = 7, and 5 = 9).

27.2. Stimuli and apparatus

The stimuli were the same as in Experiment 4.

27.3. Procedure and design

The procedure and design was the same as in Experiment 4. The only difference is that the fractions were presented simultaneously rather than sequentially. The bottom panel of Fig. 15 illustrates a typical trial sequence in Experiment 5.

28. Results

28.1. Accuracy

Performance of six participants (1, 6, 7, 11, 12 and 15) was extremely poor (accuracy <0.55), and as a result they were removed

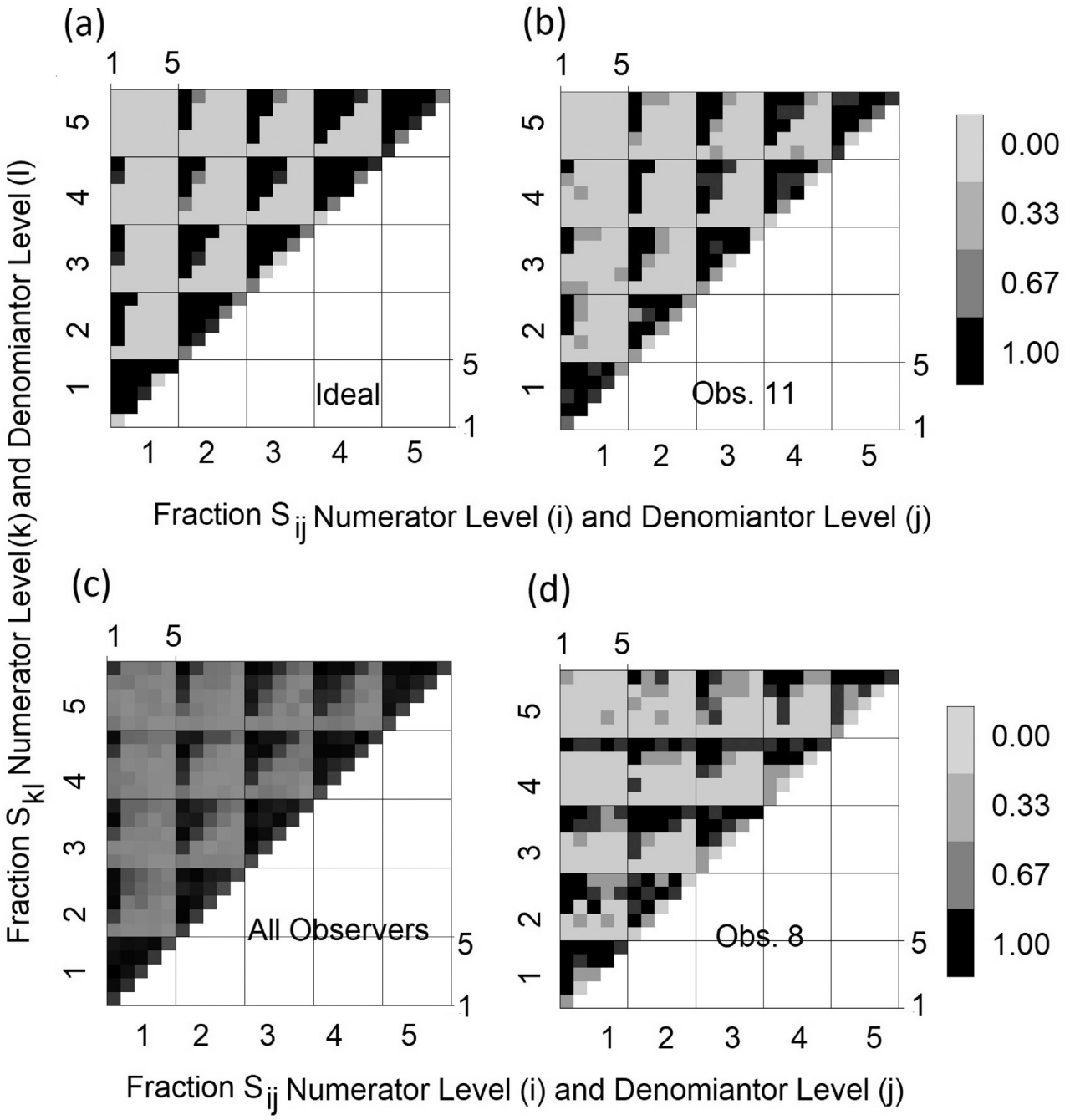


Fig. 19. Experiment 4: Conjoint proportions plots. Panel (a): Ideal-Observer (additive model), (b) Empirical results for participant 11 who conformed with the ideal-observer (additive) model, (c) Empirical results across all-participants, and (d) Empirical results for participant 8 who whose data were most consistent with a saturated observer model, which violates the ideal-observer (additive) model. Note that the numbers 1–5 in this Figure stand for the i numerator or j denominator levels and are not the actual numerator or denominator values (the mapping is 1 = 1, 2 = 3, 3 = 5, 4 = 7, and 5 = 9).

from analyses. This is not surprising given the prevalence of dyscalculia in the population and the difficulties with fractions discussed at the outset. Overall accuracy of the 18 remaining participants amounted to 79% correct ($sd = 4.07\%$). This high rate of failure to comply with the experiments demands may not come as a surprise given the fact that fractions are challenging even to educated adults (Siegler et al., 2011). Next, we assessed the presence of a distance effect by computing the absolute distance between every pair of fractions performing a linear regression with the log of accuracy and the log of absolute distance. The results [$R^2 = 0.31, F(1, 123) = 57.73, p < 0.0001$] confirmed, yet again, the presence of a global distance effect, as can be noted in Fig. 20. The finding of a distance effect with fractions (Schneider & Siegler, 2010) supports holistic processing.

28.2. MLCM

The additive and independent models were compared separately for each observer. As expected, the additive model was better fit to the data than the independent model for all participants (see Table 3).

Next, the estimated contributions of denominator and numerator magnitudes to perceived global fraction magnitude were derived. Fig. 21 illustrates representative plots of perceptual scale values as a

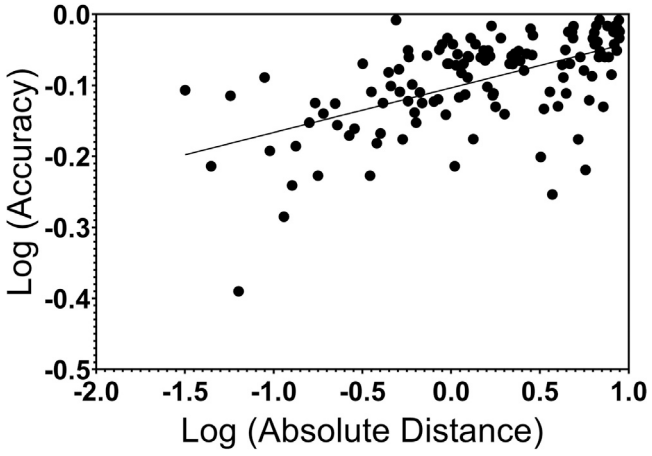


Fig. 20. Experiment 5: A distance effect for global fraction magnitude. Log Accuracy as a function of Log absolute distance.

Table 3
Experiment 5: Comparison of independent and additive observer models for judgments of fraction magnitude. The Bonferroni-corrected level for multiple comparisons is $p < .0027$. Six participants (1, 6, 7, 11, 12 and 15) were removed from analyses due to very high levels of error.

	Test fractions (additive model x independent model)		
	DF	χ^2	*p*-value
Obs. 2	4	319.43	<.001***
Obs. 3	4	343.42	<.001***
Obs. 4	4	134.94	<.001***
Obs. 5	4	71.68	<.001***
Obs. 8	4	194.19	<.001***
Obs. 9	4	224.85	<.001***
Obs. 10	4	413.56	<.001***
Obs. 13	4	52.38	<.001***
Obs. 14	4	113.06	<.001***
Obs. 16	4	199.98	<.001***
Obs. 17	4	207.47	<.001***
Obs. 18	4	524.60	<.001***
Obs. 19	4	37.85	<.001***
Obs. 20	4	213.21	<.001***
Obs. 21	4	405.45	<.001***
Obs. 22	4	415.88	<.001***
Obs. 23	4	308.04	<.001***
Obs. 24	4	489.49	<.001***

*** $p < 0.0027$.

function of the objective values, separately for numerator and denominator magnitudes. As can be noted, the overall pattern conformed with the ideal-observer pattern. However, this analysis cannot inform us on whether the integration was optimal. For this end, we need to compare the additive model with the saturated model and extract the saturated model values.

We compared the saturated and the additive models (see Table 4). At a Bonferroni-corrected level of $p < .0027$, the saturated model exhibited better fit for 14 out of 18 participants, suggesting strong violations from the additive (optimal) model. We also fit the saturated model to the data, separately for each observer data. Panel C in Fig. 22 illustrates the results at the aggregate level. These can be compared to the hypothetical results of an ideal-observer in Panel A of the same Figure. The empirical pattern is highly interactive, as can be noted by the breakdown of parallelism in the estimated functions. This strongly suggests deviations from the additive (optimal) observer model. Panel B of Fig. 22 presents the estimated parameter values for a participant who did conform with the additive (optimal) model, whereas Panel D of the same Figure, illustrates the estimated parameter values for a participant whose data were best fit by the saturated model.

Table 4
Experiment 5: Comparison of additive and saturated observer models. The Bonferroni-corrected level for multiple comparisons is $p < .0027$. Six participants (1, 6, 7, 11, 12 and 15) were removed from analyses due to very high levels of error.

	Test fractions (saturated model x additive model)		
	DF	χ^2	*p*-value
Obs. 2	16	74.19	0.001***
Obs. 3	16	18.11	0.31
Obs. 4	16	27.30	0.03
Obs. 5	16	38.57	0.001***
Obs. 8	16	63.63	0.001***
Obs. 9	16	50.95	0.001***
Obs. 10	16	28.48	0.02
Obs. 13	16	39.78	0.001***
Obs. 14	16	174.75	0.001***
Obs. 16	16	64.51	0.001***
Obs. 17	16	144.80	0.001***
Obs. 18	16	45.95	0.001***
Obs. 19	16	50.70	0.001***
Obs. 20	16	45.46	0.001***
Obs. 21	16	67.81	0.001***
Obs. 22	16	27.72	0.03
Obs. 23	16	71.79	0.001***
Obs. 24	16	40.81	0.001***

*** $p < 0.0027$.

As revealed by the model comparison procedure, the data of 14 out of 18 participants were consistent with the sub-optimal (saturated) model. This strongly suggest improper processing of fractions. Deviations from optimality can also be noticed by comparing Panel A and Panel D in Fig. 23 which plots the response proportion (Ho et al., 2008; Knoblauch & Maloney, 2012) for an ideal-observer and for a representative participant (participant 24) whose data violated the optimal (additive) model. The reader is invited to compare the aggregate pattern (Panel C), with that of participant 3 (Panel B), whose data conformed to the ideal-observer.

29. Discussion

Experiment 5 provides novel insights into the internal representations that guide fractions processing. In this experiment pairs of fractions were presented simultaneously rather than sequentially for the task of comparative judgment. Of the 18 participants, 14 exhibited violations of the optimal (additive) strategy. This rate of 78% is remarkable, given that in Experiment 4, which presented the fractions in a sequential mode, only 9 out of 18 (50%) participants showed this trend. Taken together, Experiment 4 and 5 suggest that educated adults fail to process the fraction's denominator and numerator in an optimal fashion. We also documented a global distance effect, which supports a holistic representation, but the exact nature of this representation could be exposed only by understanding how the components are integrated. This has been achieved by applying the MLCM methodology. The results reveal that in comparative judgments of fractions most educated adults cannot represent the fractions in the correct fashion. They can still achieve accuracy level of around 80%, but this is not because they accurately represented the fractions, but likely due to the ability to approximate the respective magnitudes (Kallai & Tzelgov, 2009). This supports the notion that, for most educated adults, the comparison of fractions is governed by fast and frugal heuristics rather than by precise algorithmic computation or an accurate analog representation.

30. General discussion

Many children (Bright et al., 1988; Hartnett & Gelman, 1998; Siegler et al., 2011) and adults (McCloskey, 2007) experience difficulties with fractions. A robust tendency to use single-unit counting to interpret

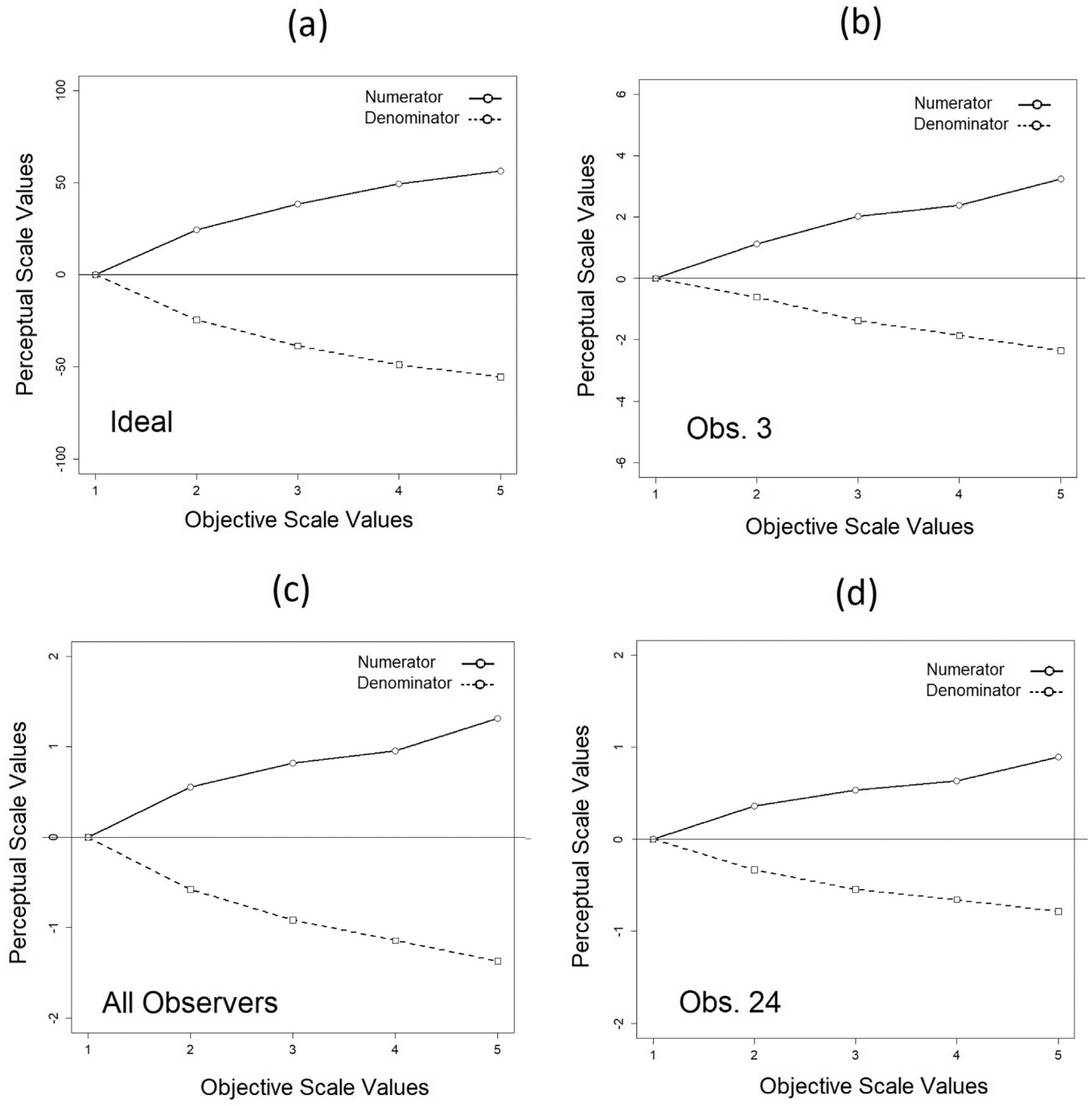


Fig. 21. Experiment 5: Estimated perceptual scales as a function of objective values obtained by fitting the additive MLCM model. Panel (a): Ideal Observer (additive model) pattern, (b) Participant 3 whose data conformed with an ideal observer, (c) all observers, and (d) Participant 24 whose data violated the ideal observer model. Note that the numbers 1–5 in this Figure stand for the *i* numerator or *j* denominator levels and are not the actual numerator or denominator values (the mapping is 1 = 1, 2 = 3, 3 = 5, 4 = 7, and 5 = 9).

fractions – "whole number bias" (Ni & Zhou, 2005) – likely compromises their ability to represent and manipulate fractions accurately as performing adding or multiplying fractions (Siegler et al., 2013). In their "integrated theory of numerical development" Siegler et al. (2011) argued that acquisition of fractions requires an overhaul reorganization of knowledge to accomplish the ability to map different types of numbers onto a number-line. This developmental stage may not be successfully attained even by educated adults. However, to-date an investigation into the internal representations and the integration processes that determines fractions representational precision has not be conducted. In this study, we sought to uncover the internal representation that guide educated adults in various canonical tasks with fractions. We focused on the precision of the representation and the operative combination rule by which observer combined denominators and numerators onto global magnitude.

Five experiments were conducted with proper and improper symbolic fractions, written mostly in Arabic notation. The first three experiments deployed a number-to-line task (Siegler et al., 2009), whereas the last two experiments administrated a non-speeded comparative judgment task (Moyer & Landauer, 1967). The construction of the stimuli set in all experiments was carefully planned to provide a factorial or semi factorial design. These stimuli sets allowed us to perform rigorous model testings of the internal representations that guide educated adults in understanding and manipulating fractions. We tested hypotheses with respect to both the global magnitude, and the separate

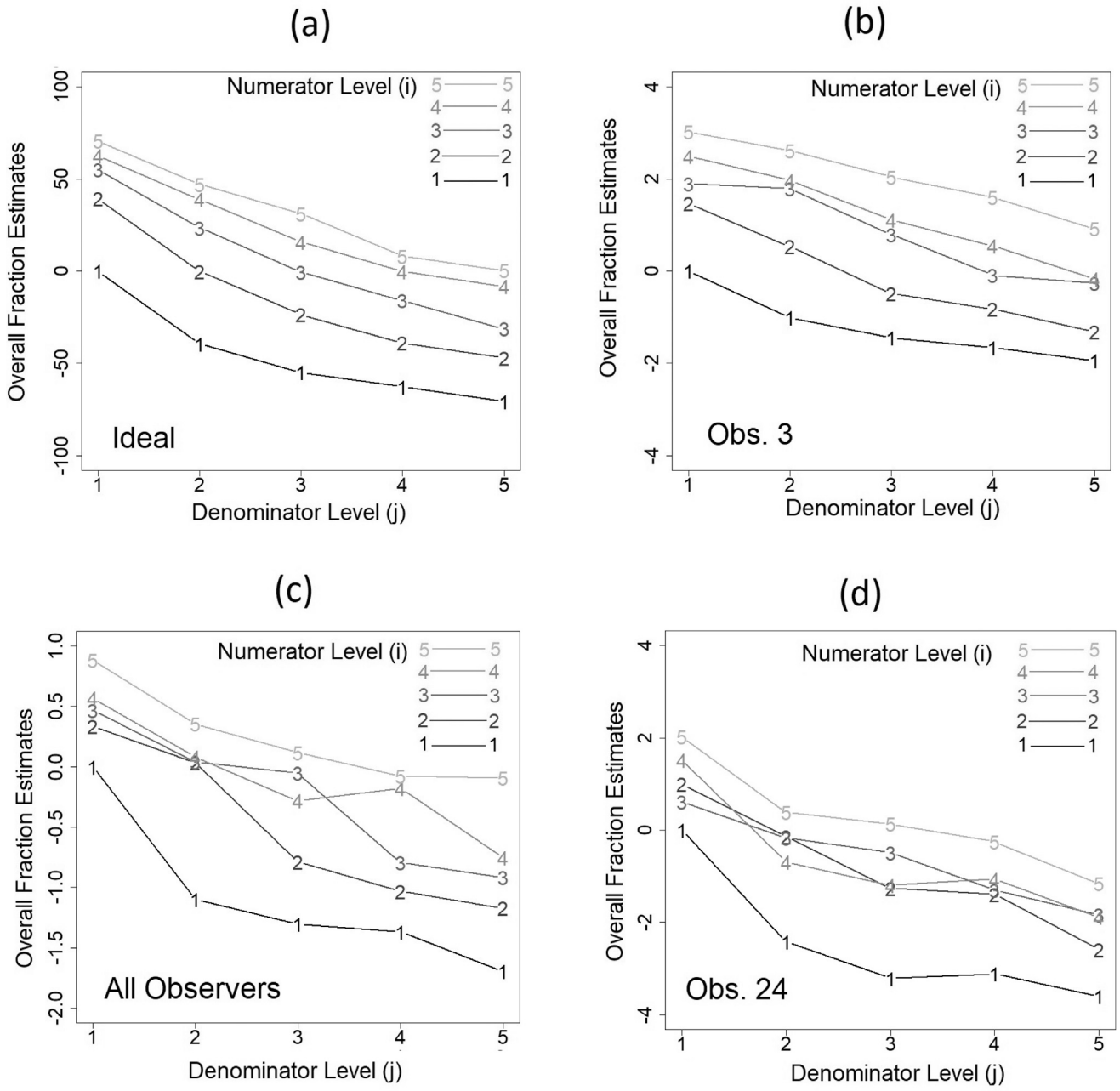


Fig. 22. Experiment 5: Overall fraction estimates as a function of denominator level for each level of numerator. Panel (a): Ideal-Observer (additive model), (b) Empirical results for participant 3 who conformed with the ideal-observer (additive) model, (c) Empirical results across all-participants, and (d) Empirical results for participant 24 whose data were most consistent with a saturated observer model, which violates the ideal-observer (additive) model. Note that the numbers 1–5 in this Figure stand for the *i* numerator or *j* denominator levels and are not the actual numerator or denominator values (the mapping is 1 = 1, 2 = 3, 3 = 5, 4 = 7, and 5 = 9).

contributions of denominators and numerators. We compared human performance in these tasks to that of an ideal-observer using functional measurement (Anderson, 1981) and conjoint measurement (Luce & Tukey, 1964) techniques.

Experiments 1–3 administrated a number-to-line task. In Experiment 1 we presented proper fractions in Arabic notation. We found evidence for a linear psychophysical function with scalar variability. Functional measurement analysis suggested that denominator and numerator were combined according to either a holistic-linear or analytic-linear integration rule (the two are indistinguishable mathematically because they mimic each other). Experiment 2 presented the same proper fractions, but in number-word format. The results were comparable, except for the absence of scalar variability. In Experiment 3, both proper and improper fractions were presented. The results were similar to those observed in Experiment 1. Experiments 4 and 5 administrated a non-speeded comparative judgment task with proper and improper fractions. These were written in Arabic notation and presented in either a serial or simultaneous mode. MLCM models were fit to these data. The results indicated that the great majority of participants combined the fractions components according to a saturated (interactive) observer model, that is inconsistent with an ideal-observer (additive) MLCM model. In addition, a distance effect was found in both experiments.

31. A two-systems interpretation

One immediate conclusion that can be drawn from our five experiments is that the processing of fractions is governed by, not one, but several types of internal representations, and that the operative

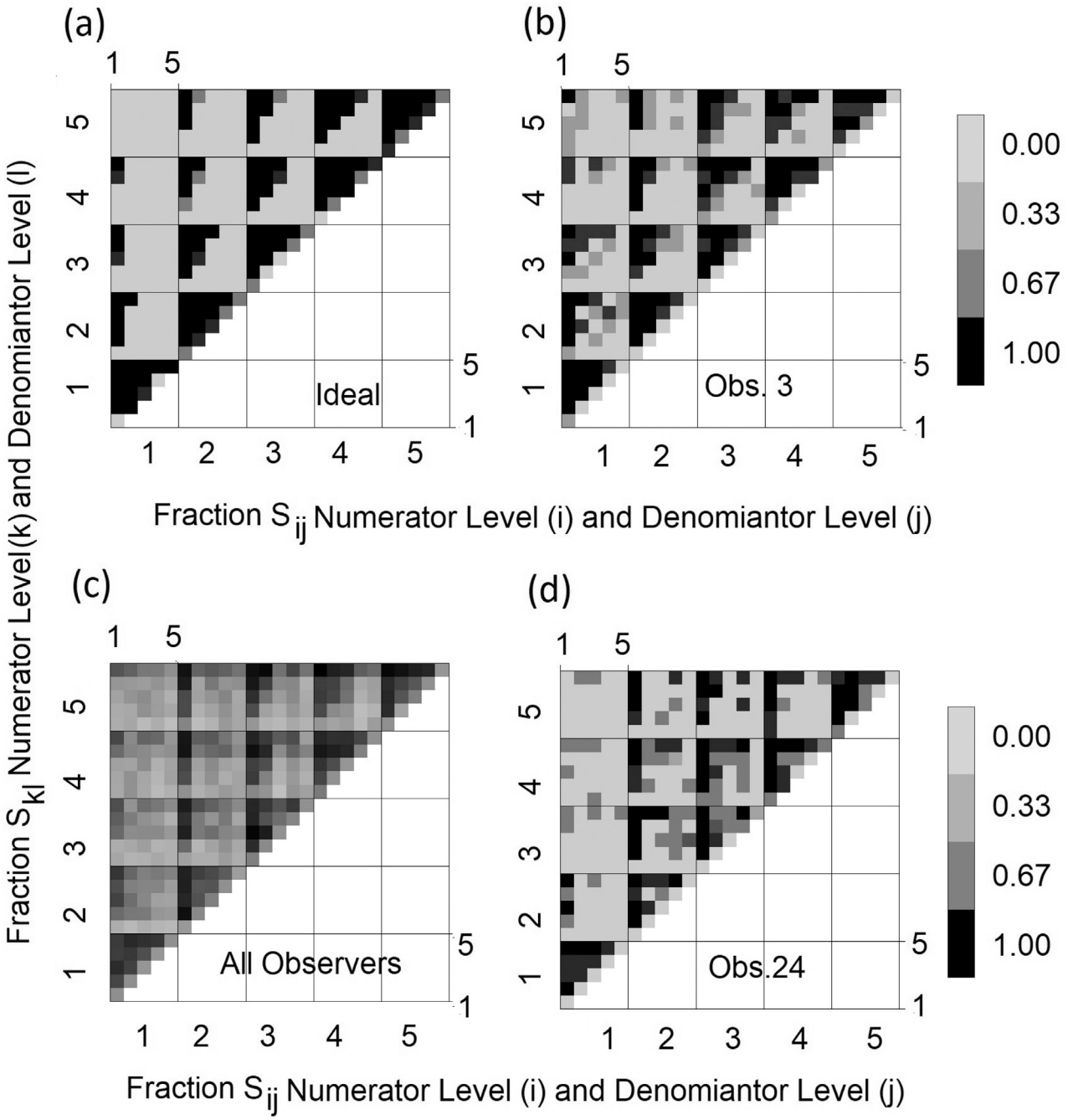


Fig. 23. Experiment 5: Conjoint proportions plots. Panel (a): Ideal-Observer (additive model), (b) Empirical results for participant 3 who conformed with the ideal-observer (additive) model, (c) Empirical results across all-participants, and (d) Empirical results for participant 24 whose data were most consistent with a saturated observer model, which violates the ideal-observer (additive) model. Note that the numbers 1–5 in this Figure stand for the *i* numerator or *j* denominator levels and are not the actual numerator or denominator values (the mapping is 1 = 1, 2 = 3, 3 = 5, 4 = 7, and 5 = 9).

representation is task dependent. We found that fractions are processed differently in comparative judgment (Experiment 4 and Experiment 5) and in number-to-line tasks (Experiment 1–3). The comparative judgment task was mostly preformed in a sub-optimal fashion according to inadequate combination rules and with higher rates of errors; whereas the number-to-line task was handled quite accurately according to normative combination rules. Why is it so? One plausible explanation is that the two tasks engage different processing systems (Evans & Stanovich, 2013; Kahneman, 2011). The comparative judgment task likely recruits the so called "system one" – a more primitive fast, automatic and inaccurate system that relies on heuristics and approximations (Tversky & Kahneman, 1974); whereas the number-to-line task is governed by the so-called "system two" – a more sophisticated, controlled system that operates according to the normative rules of arithmetic.

In addition, the comparative judgment task exerts larger cognitive load on participants as it requires the processing of two fractions rather than one. This difficulty may facilitate the operation of system-one, which in the domain of numerical-cognition is often dubbed the "approximate-number-system" (Dehaene et al., 1998). The approximate number system relies on gross and inaccurate representations, and not on accurate computational algorithm. This can explain the finding of a modicum level of success (70%−80%) that is often been documented in the comparative judgment task. The conjecture is also supported by

the findings that a sequential mode of fractions' presentation (Experiment 4) led to lower error rates and a lower number of participants who exhibited sub-optimal performance than did a simultaneous mode of presentation (Experiment 5). Simultaneous presentation is often related to limited processing capacity (Shiffrin & Gardner, 1972). The number-to-line task is more of a perceptual task, and as such, it can be performed with more precision. Moreover, the perceptual basis of this task enhances the engagement of an elaborated "computational" system, which provides more accurate estimations, probably through algorithmic computations (Logan, 1988). However, it should be noted that although performance in the number-to-line task conformed to a linear (veridical) transformation, it was subjected to scalar variability. This pattern is comparable to the one found in educated adults' performance with integers (Siegler & Opfer, 2003). Scalar variability is not fully consistent with optimal performance because it entails noisy representation, which may serve as another contributing factor to the difficulties people are experiencing with fractions.

We found that scalar variability was absent when fractions were presented in a word-format (Experiment 2). This result is remarkable because it reveals that differences in format can alter the internal representations of fractions. It is likely that the Hebrew words of fractions make the meaning of fractions more transparent than the routine Arabic notation, and as a result allow for a more precise representation. Several studies have shown that linguistic differences can affect the proficiency by which learners use fractions (Miura et al., 1999; Paik & Mix, 2003). For example, the word for 1/3 in East Asian languages is "three parts, one", in contrast to English and Hebrew phrases (one third). The East Asian phrase facilitates the part-whole meaning of the fraction, whereas both the English and Hebrew terms do not. This may explain why East Asian second graders outperform their US peers in a task that requires shading squares within a matrix to match a written fraction. Comparable differences may exist between word and Arabic notation. Notably, even in Hebrew or English, the word notation makes the part-whole relations more transparent than the Arabic notation.

32. A wave theory interpretation

The existence of multiple types of representations of symbolic fractions is consistent with the overlapping waves theory (Siegler, 1996; Siegler & Opfer, 2003; Siegler et al., 2011). The theory postulates that people hold in their memory various strategies for solving a problem. The type of strategy that will be used on a given occasion depends on the person and on the task at hand. In the number-to-line task, people used an optimal composition rule. However, in the comparative judgment task, the majority of participants employed a sub-optimal strategy, which manifested more strongly in the simultaneous than in the serial presentation mode. Moreover, the amount of noise in the number-to-line estimation scaled with global magnitude for Arabic notations but not for word notation. This finding also supports the conclusion that multiple types of fraction representations exist. All of these entails that when dealing with fractions, people can use different types of representations and strategies depending on the task at hand. The number-to-line task is easier because it relies on visual aspects, and thus allows the retrieval of more precise representations; whereas the comparative judgment task is a more abstract and difficult one. The wave theory can also explain the finding that the number of participants who deployed an optimal composition rule decreased as the fractions presentation shifted from sequential (Experiment 4) to simultaneous (Experiment 5) mode. This is likely due to the increase in difficulty.

33. Fraction: Moving beyond the holistic vs. Analytic question

We have argued that the question of whether fractions are represented holistically – as integrated wholes (DeWolf et al., 2014), or analytically – as two separate components (Bonato et al., 2007; Kallai & Tzelgov, 2009), is secondary to the question of the actual precision of representation. We are not arguing that the question is inconsequential nor that it has nothing to do with the second question. But we stress the point that in many cases, holistic and analytic representations cannot be experimentally dissociated due to the problem of mimicry. We have demonstrated this point by showing that mathematically, the two types of representations can mimic each other when translated to behavioral effects. For example, in the context of the functional measurement approach (Anderson, 1981), linear-holistic and linear-analytic models predict the same number-to-line function. Moreover, it is tempting to argue that a linear representation with scalar variability is generated by a holistic representation. However, scalar variability can also result from motor noise that is added to an analytic representation. Even when the components strongly interact with each other, as we have shown in the MLCM models of the comparative judgment task (Experiment 4–5), one can still argue that this reflects a noisy long-term internalized holistic representation. The upshot is that it is almost impossible to tease a part the internal representations and determine their status as holistic or analytic. What is more important is to determine how the final representation (holistic or analytic) is composed of the basic elements, and whether participants follow the normative rules of this construction. This is exactly what we sought to accomplish in this study.

34. Why fractions are and are not difficult?'

We commenced our endeavor by asking why so many children and educated adults find fractions difficult to manipulate and understand. The traditional answer is that people are biased toward processing fractions as if they were integers (Ni & Zhou, 2005). People tacitly assume that fractions are governed by the same mathematical rules as integers. Integers are written as a single symbol, have unique successors, are countable, and never increase by division (Siegler et al., 2011). Fractions, in contrast, do not obey all of these rules. In a matter of fact, fractions are characterized by unique mathematical operations among their components. For example, adding two fractions requires finding a common denominator, an operation that is not needed in integer arithmetic. Moreover, as the magnitude of the denominator increases the global magnitude of the fraction gets smaller. Some people find these inverse relations confusing. A fraction, by-definition, requires dividing attention between two symbols. Even if the ultimate representation is holistic, it still undergoes an initial construction from two separate symbols. The goal of any successful educational or developmental trajectory from integers to fractions is to understand the consequences of division. Our investigation addressed this ultimate source of difficulty. In this study we show a lacuna in the traditional view. Fractions can be easy in some tasks, but difficult in others. This depends on the task at hand and on the type of internal representations retrieved from memory or the system (one or two) that is deployed to tackle the problem. We have tested for semantic knowledge of fractions, and identified two major sources of difficulty that can hamper the way procedural knowledge is handled: (a) representations are not accurate due to scalar variability, and (b) representations are not precise due to application of improper integration rules. We suggest that educators should put more effort in addressing these issues to help learners become proficient users of fractions.

35. Limitations and future directions

The present study applied rigorous methods to uncover the underlying composite rules by which numerators and denominators are integrated by educated adults. However, there are several limitations to our endeavor. First, the range of fractions used was limited to the 0–1 and 0–10 range, with denominators and numerators ranging only between 0–9. Future work should extend the application of the present methods to a broader range of fractions. Second, the participants in

our study were all psychology students. It would be interesting to compare their results with those of students from natural sciences, and see whether similar failures in applying the additive model (as observed in Experiment 4 and 5) characterize more mathematically-oriented participants. Second, we did not screen our participants for dyscalculia. It might be the case that our sample included some dyscaluculic students. In addition, it might be valuable in future research to relate individual differences in fractions representations to individual differences in other mathematical abilities using our methods. Third, we did not collect verbal reports by participants regarding their strategy usage. In future work, it would be beneficial to relate our quantitative measures to participants' own self-knowledge of their strategies. Finally, there is a great potential in importing the conjoint measurement methods used here to developmental research. These methods are simple to implement, are not speeded, are based on order scales, and can provide new insights into the acquisition of fractions by children.

CRediT authorship contribution statement

Daniel Fitousi: Wrote the paper, Designed the experiments, Conceived the experiments. **Ran Noyman:** Run the experiments, Conceived the experiments.

Data availability

Data can be downloaded from https://data.mendeley.com/datasets/zd6wntrxdc.

Acknowledgments

Part of this study (Experiment 1 and Experiment 2) was reported in a thesis submitted to Ariel University by the second author under the supervision of the first author.

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Open practice statement

All data, codes, and supporting materials can be downloaded from here Data and codes can be downloaded from https://data.mendeley.com/datasets/zd6wntrxdc

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